Implements

Overview of all implements for the Bruut OpenAgbot: the 15 that are worked out in FreeCAD, with every variant on its own page, followed by the 31 ideas from the website that are not worked out yet.

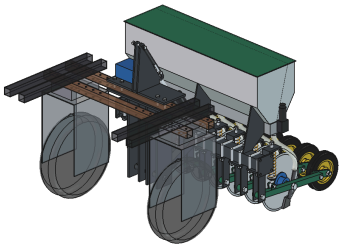
Contents

1. **Overseeder**
   Sowing & Planting
2. **Liquid fertilizer applicator**
   Spraying, Fertilizing & Irrigation
3. **Feed pusher auger**
   Transport & General
4. **Disc brushes**
   Transport & General
5. **Dockweed drill**
   Soil Cultivation & Weed Control
6. **Trencher**
   Soil Cultivation & Weed Control
7. **Field marker**
   Measuring & Sensors
8. **Mobile refiller**
   Spraying, Fertilizing & Irrigation
9. **Around-pole fence mower**
   Mowing
10. **Weed picker**
    Mowing
11. **Irrigation**
    Spraying, Fertilizing & Irrigation
12. **Spot sprayer**
    Soil Cultivation & Weed Control
13. **In-row weeder**
    Soil Cultivation & Weed Control
14. **Belt rake**
    Soil Cultivation & Weed Control
15. **Weeds burner (flame weeder)**
    Soil Cultivation & Weed Control
16. **Implement ideas (not worked out yet)**

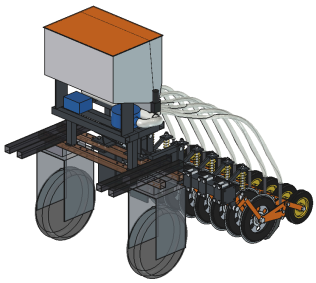
IMPLEMENT · SOWING & PLANTING

Overseeder

Grassland slit seeder. 1 m modules with 8 rows at 125 mm and coupling flanges. A thin disc cuts a 15 mm slot, the seed is placed about 12 mm deep and a sprung press wheel closes the slot. Every row follows the ground on its own arm; two gas springs push the floating lift frame down with robot weight.



Simple

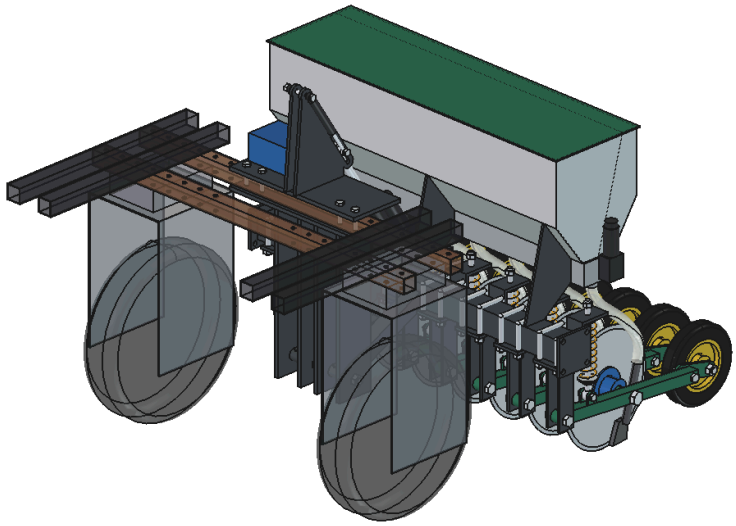


Advanced

2 variants: Simple · Advanced
Folder: `overseeder/`

OVERSEEDER · VARIANT 1 OF 2

Simple



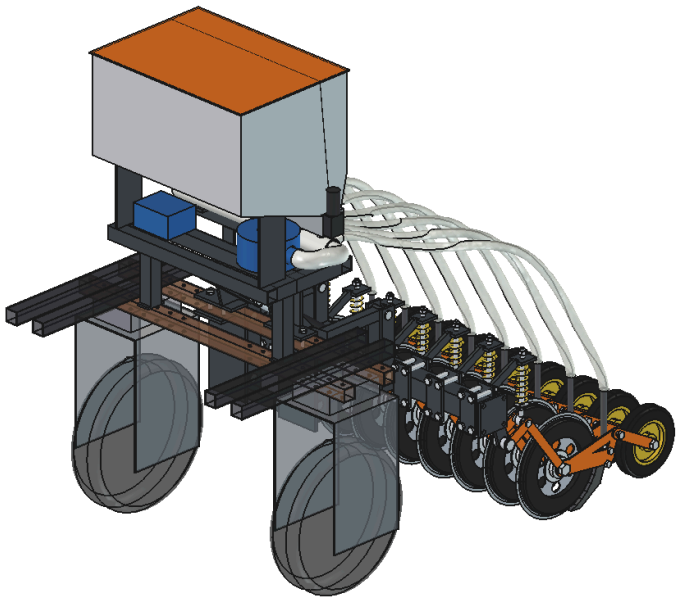
Angled single disc (7°, Ø300 × 3) with a seed boot in its lee and a depth ring. Gravity-fed seed hopper with a compartment for grass (41 l) and one for fine seed (18 l), each with its own cam roller and a motor that follows the driving speed. Built on the headstock and lift frame of the simple applicator v2.

- 8 rows at 125 mm = 1 m per module
- Draft force approx. 335 N (normal)
- Depth within ±3 mm: 94 %
- About €1,675 in parts

Folder: `overseeder/simple/`

OVERSEEDER · VARIANT 2 OF 2

Advanced



Parallel linkage so the toolbar stays parallel, straight Ø300 × 3 disc with depth rings on both sides, a curved seed coulter directly behind the disc and a press wheel in a fork. Air seeder (62 + 17 l hopper, 12 V fan) on a frame above the rear axle blows the seed through 8 hoses to the coulters.

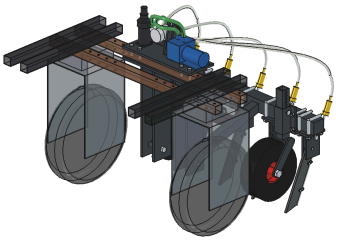
- 8 rows at 125 mm = 1 m per module
- 2 gas springs in the slotted hole
- Depth within ±3 mm: 98 %
- About €2,280 in parts

Folder: `overseeder/advanced/`

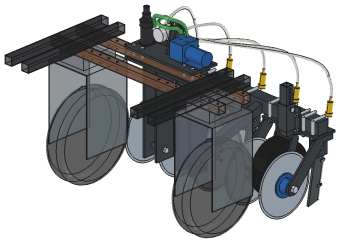
IMPLEMENT · SPRAYING, FERTILIZING & IRRIGATION

Liquid fertilizer applicator

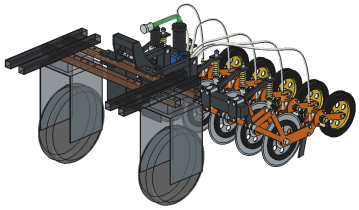
Injects liquid fertilizer (such as RENURE) just under the sward, without mist or drift. Five injection elements at 200 mm; each element follows the ground on its own arm.



Simple



Simple v2

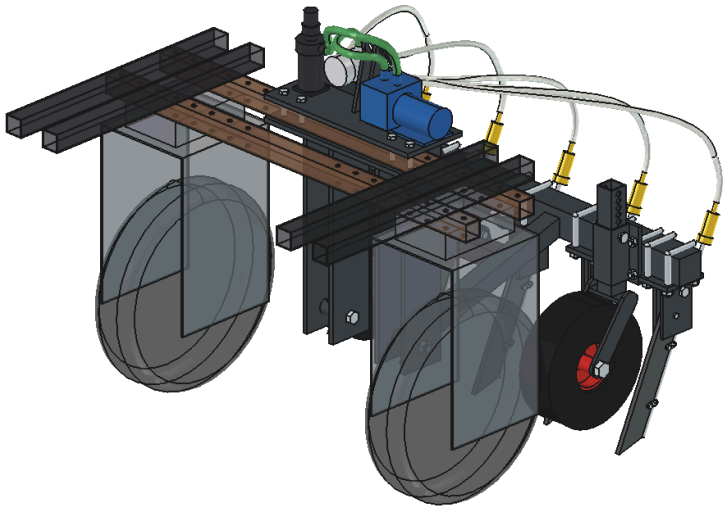


Advanced

3 variants: Simple · Simple v2 · Advanced
Folder: `liquid fertilizer applicator/`

LIQUID FERTILIZER APPLICATOR · VARIANT 1 OF 3

Simple



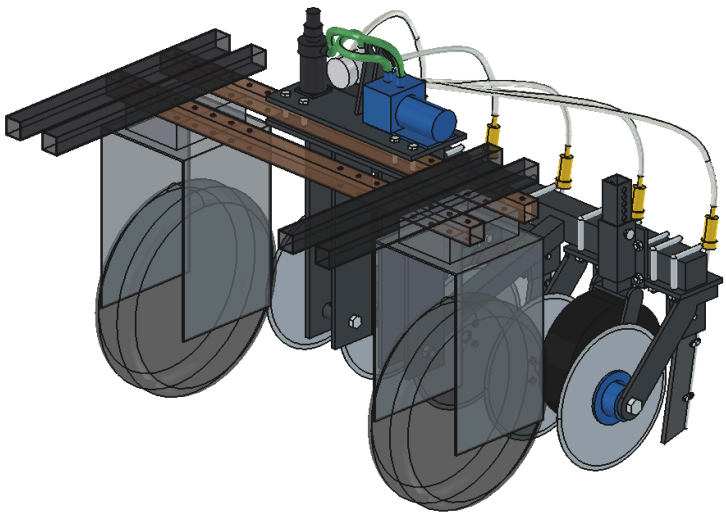
Cheap and robust: 5 fixed injection knives with a stainless steel tube, two wheelbarrow wheels as depth wheels, a lift frame with a single low pivot and one cheap linear actuator, a 12 V diaphragm pump with fixed regulator and orifice plates. Only flat bar, tube and standard parts.

- 5 knives at 200 mm
- 12 V diaphragm pump, dose depends on speed
- About €600 in materials

Folder: `liquid fertilizer applicator/simple/`

LIQUID FERTILIZER APPLICATOR · VARIANT 2 OF 3

Simple v2



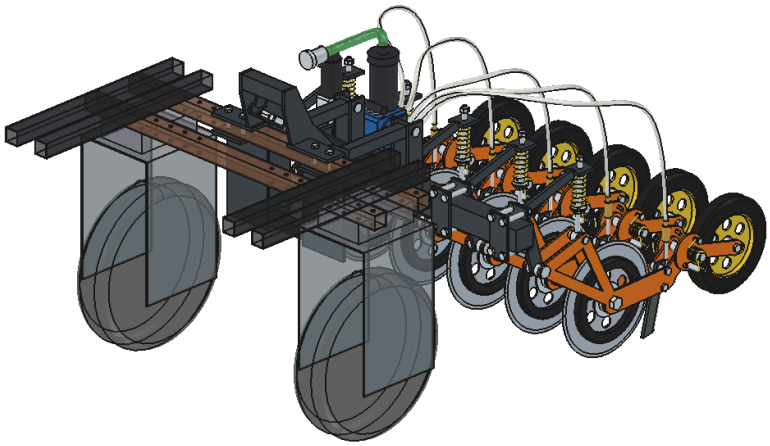
Simple version plus a flat coulter disc Ø300 in front of each knife, cutting 45 mm deep; the knife runs 14 mm behind it in the same plane. Costs 26 kg extra on the lift frame (70 kg), a 1500 N actuator and up to 50 kg of ballast in hard sward.

- Disc + knife per row
- Lift frame 70 kg
- About €915 in materials

Folder: `liquid fertilizer applicator/simple_v2/`

LIQUID FERTILIZER APPLICATOR · VARIANT 3 OF 3

Advanced



Disc, knife, tube and press wheel per element on spring arms; a 5-channel peristaltic pump driven by a worm-gear motor whose speed follows the robot's driving speed, and an electrically lifted parallel linkage that floats while working (slotted hole). No ground wheel.

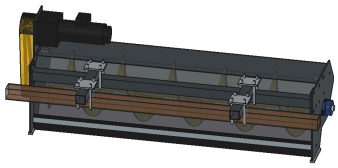
- 5 elements at 200 mm
- 89–1,316 l/ha, 18 settings
- Includes ground-following animation

Folder: `liquid fertilizer applicator/advanced/`

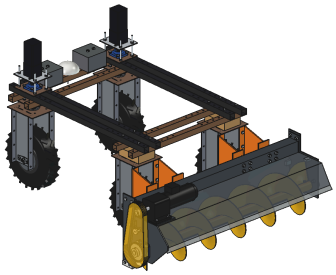
IMPLEMENT · TRANSPORT & GENERAL

Feed pusher auger

Pushes feed back to the feed fence. Sits at the back of the robot; the robot drives with the auger first. The auger picks up the pushed-away feed and carries it sideways to the fence.



Simple

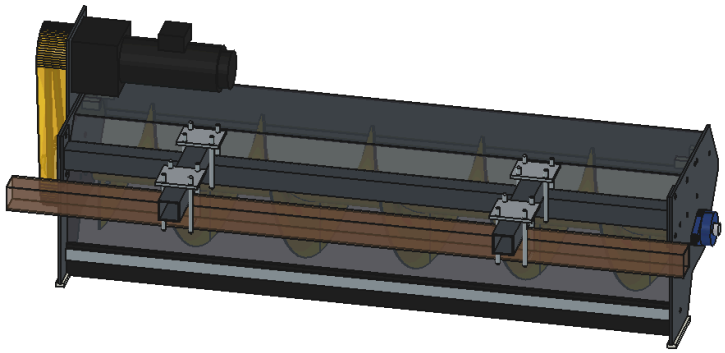


Advanced

2 variants: Simple · Advanced
Folder: `feed pusher auger/`

FEED PUSHER AUGER · VARIANT 1 OF 2

Simple



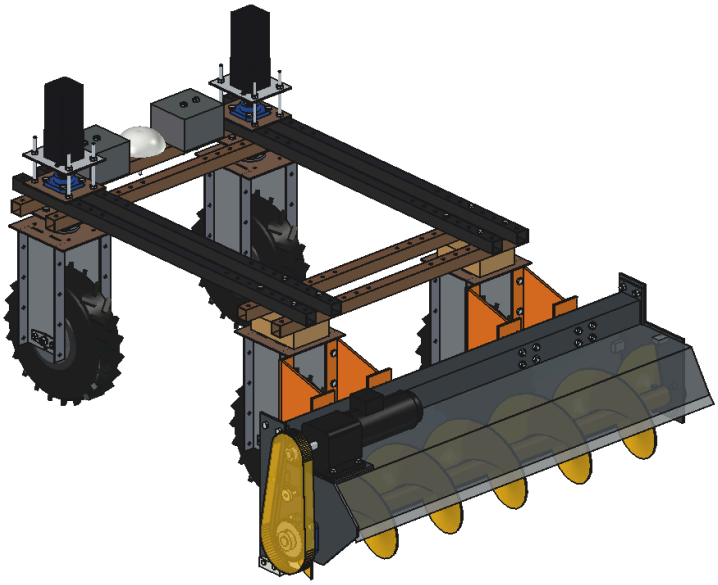
Cheap version from flat plate and sectional flights; DXF files for the side plate and the flight ring.

Flat plate + sectional flights	STEP and DXF files

Folder: `feed pusher auger/simple/`

FEED PUSHER AUGER · VARIANT 2 OF 2

Advanced



Ø320 auger under a 2 mm folded hood with 4 bends. Foot-mounted gearmotor on top of the hood, 08B-1 chain with tensioner along the left side plate. Mounted with 4 arms on the rear flange holes of the rear wheel modules. Open discharge end. Includes a feed-transport model and a force animation.

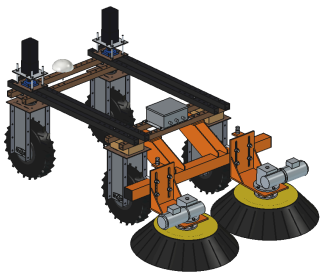
- 24 V gearmotor with chain drive
- Open discharge end
- Feed-transport model and force animation

Folder: `feed pusher auger/advanced/`

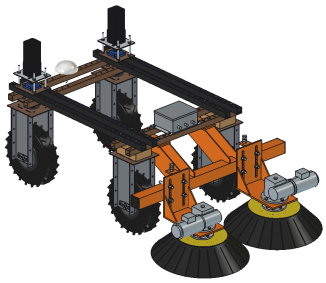
IMPLEMENT · TRANSPORT & GENERAL

Disc brushes

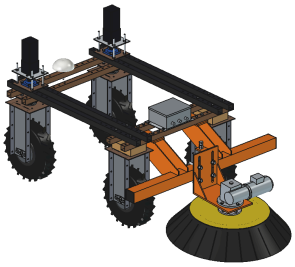
Sweeper disc brushes for loose material on concrete. Each brush on its own 48 V worm gearmotor (NMRV063, 550 W); reversing the motors makes a windrow in the middle or spreads material wide. Mounted on an 80 × 80 toolbar under the rear beams with U-bolts. Includes an animation with a particle model.



Double 750



Double 650

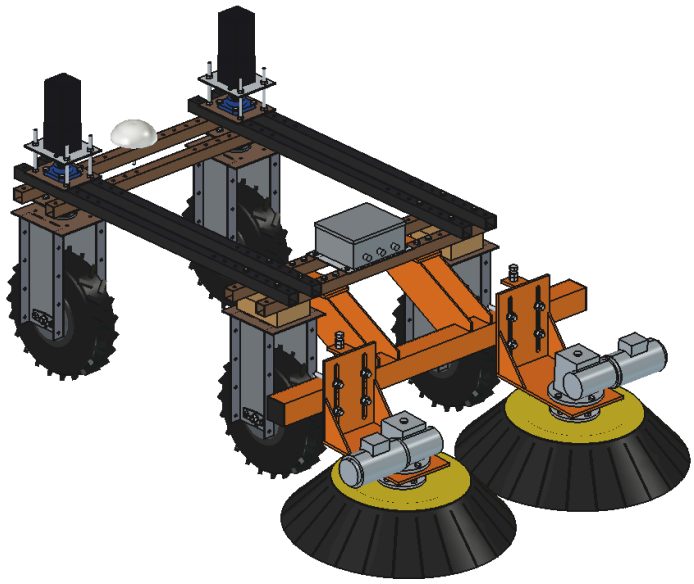


Single 850

3 variants: Double 750 · Double 650 · Single 850
Folder: `brushes/`

DISC BRUSHES · VARIANT 1 OF 3

Double 750



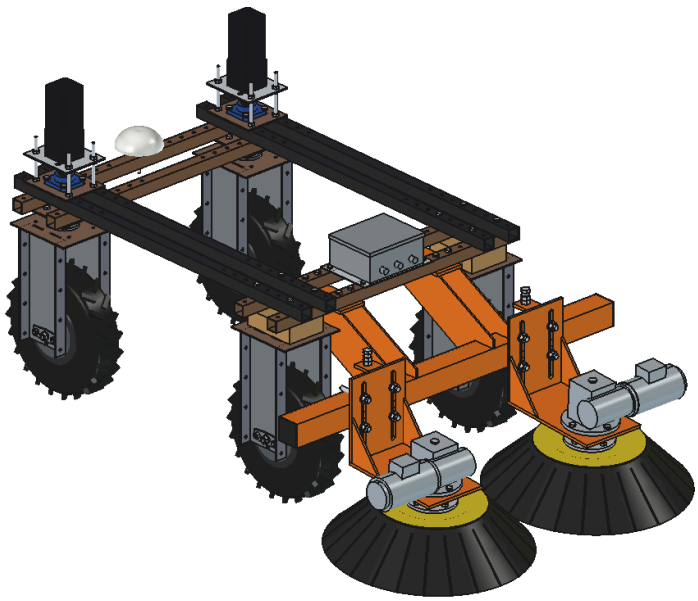
Two Ø750 brushes (standard). Working width 1.50 m with brushes touching to 1.70 m with 200 mm between them. Spread wide or make a windrow.

2 × Ø750	1.50–1.70 m

Folder: `brushes/simple/`

DISC BRUSHES · VARIANT 2 OF 3

Double 650



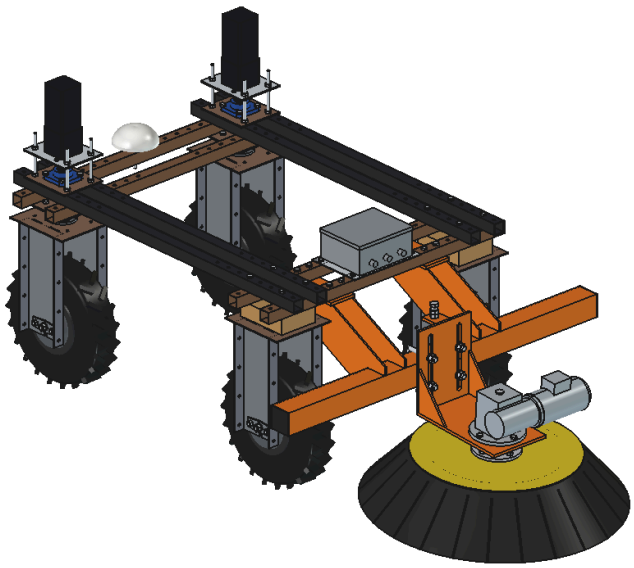
Two Ø650 brushes: the same job, narrower and lighter.

2 × Ø650	1.30–1.50 m

Folder: `brushes/simple/`

DISC BRUSHES · VARIANT 3 OF 3

Single 850



One Ø850 brush covering the robot's own width, sweeping the strip in front of the robot to one side.

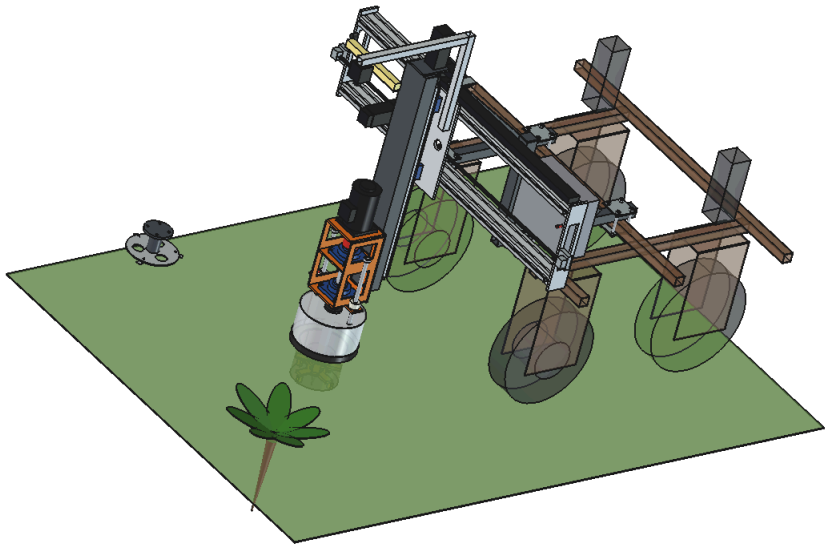
1 × Ø850	0.85 m

Folder: `brushes/simple/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Dockweed drill

Finds a dock plant (broad-leaved dock) and mills out its root on a CNC gantry, chemical-free.

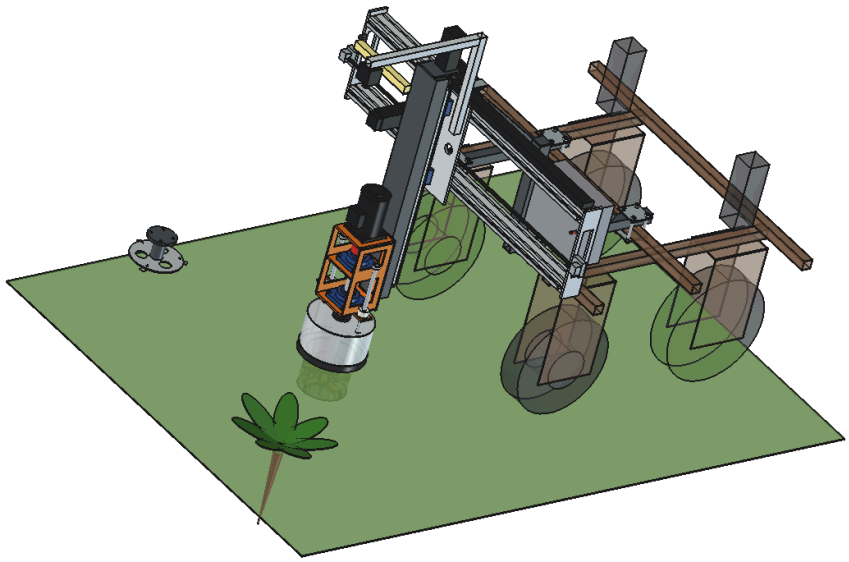


One design

1 variant: One design
Folder: `dockweed drill/`

DOCKWEED DRILL · VARIANT 1 OF 1

One design



Dock removal mill on a two-axis CNC gantry. Renders, animations, DXF and STEP files. An older single-script design (dockweed_design.py).

> Animation and close-up GIFs

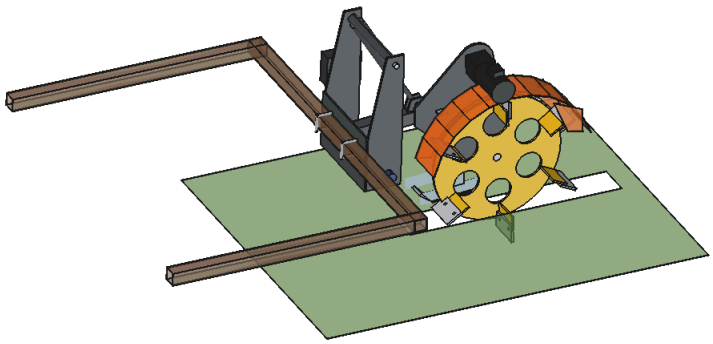
> DXF and STEP files

Folder: `dockweed drill/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Trencher

Wheel trencher for cutting narrow trenches.

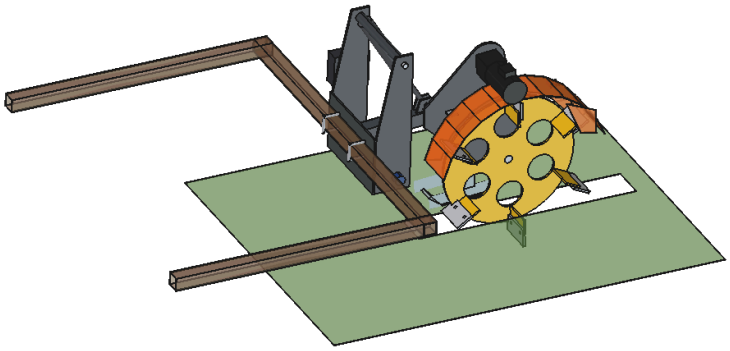


One design

1 variant: One design
Folder: `trencher/`

TRENCHER · VARIANT 1 OF 1

One design



Wheel trencher mounted on the Bruut. Renders and STEP export. An older single-script design (trencher_design.py).

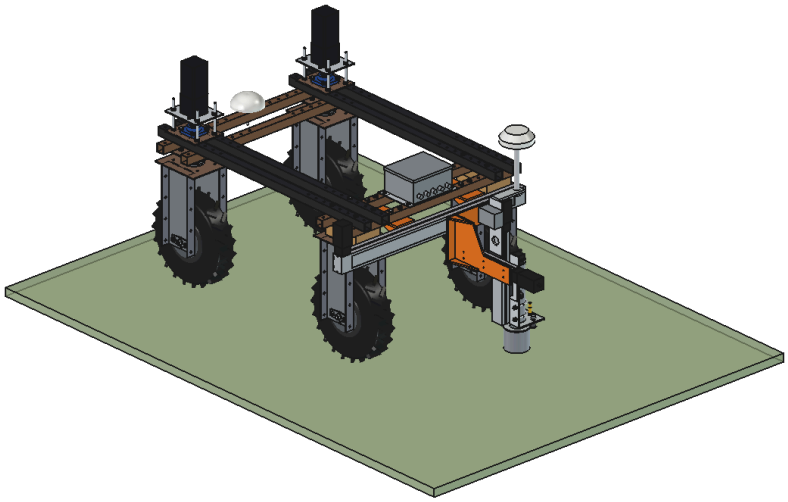
> Renders and STEP export

Folder: `trencher/`

IMPLEMENT · MEASURING & SENSORS

Field marker

Stakes out points from a coordinate list (piles, posts, planting positions, plot corners). The robot stops roughly behind the point; an XYZ stage puts a spray-paint head exactly on it.

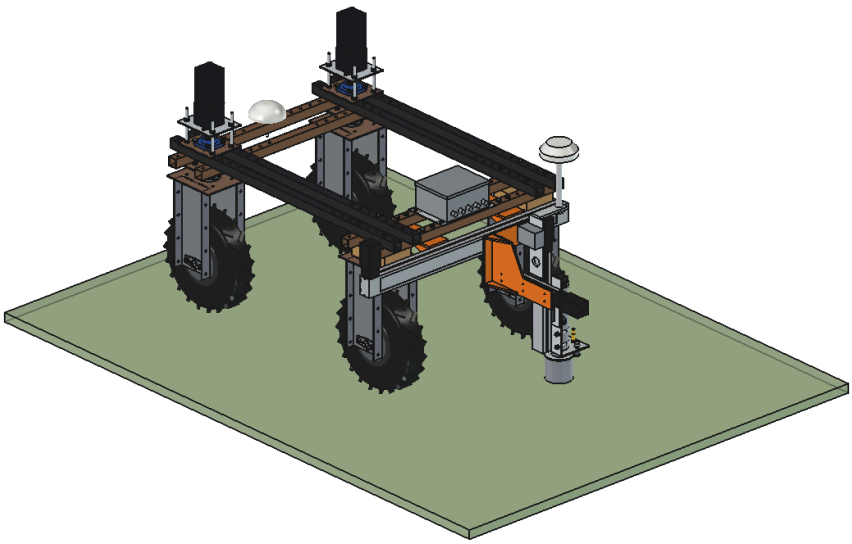


Simple

1 variant: Simple
Folder: `marker/`

FIELD MARKER · VARIANT 1 OF 1

Simple



Three bought linear modules: X 800 mm belt module, Y 200 mm and Z 300 mm ball screws. Paint head: upside-down marking-spray can in a PVC drain pipe pushed by a servo; sprung PVC 110 shroud with an inductive sensor finds the ground. Second GNSS antenna on a mast straight above the nozzle.

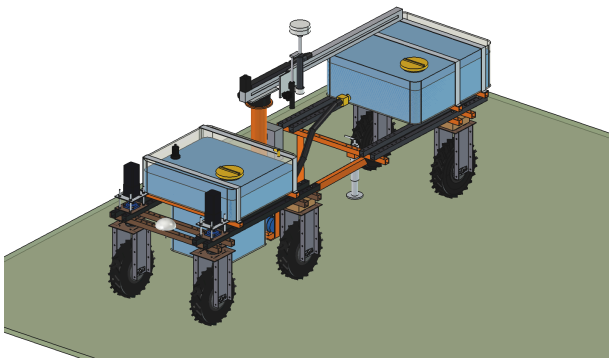
- Robot may stop ±100 mm off
- NEMA 23 closed-loop motors
- About €1,450 in parts
- Concept lists flag, peg, punch and soil-probe heads

Folder: `marker/simple/`

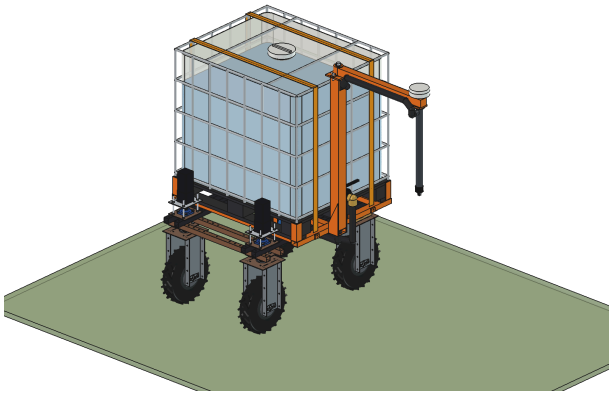
IMPLEMENT · SPRAYING, FERTILIZING & IRRIGATION

Mobile refiller

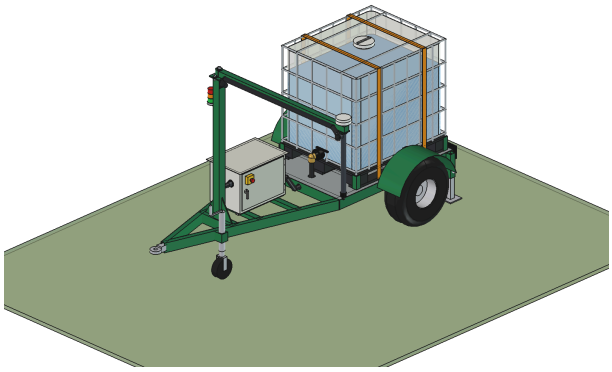
Support robot that brings liquid (and in concept seed, granular fertilizer, water, batteries) to the working robots, so they never leave the field.



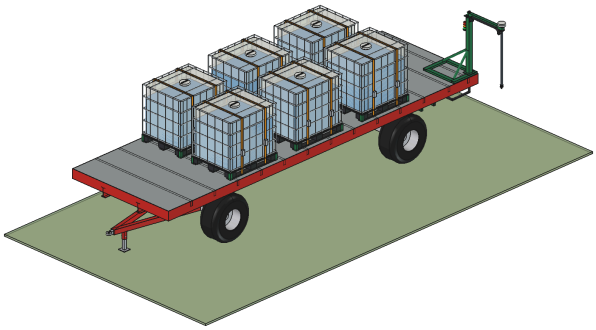
Liquid



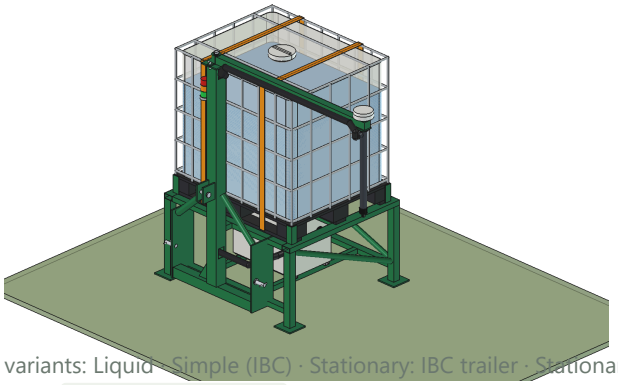
Simple (IBC)



Stationary: IBC trailer



Stationary: flat wagon



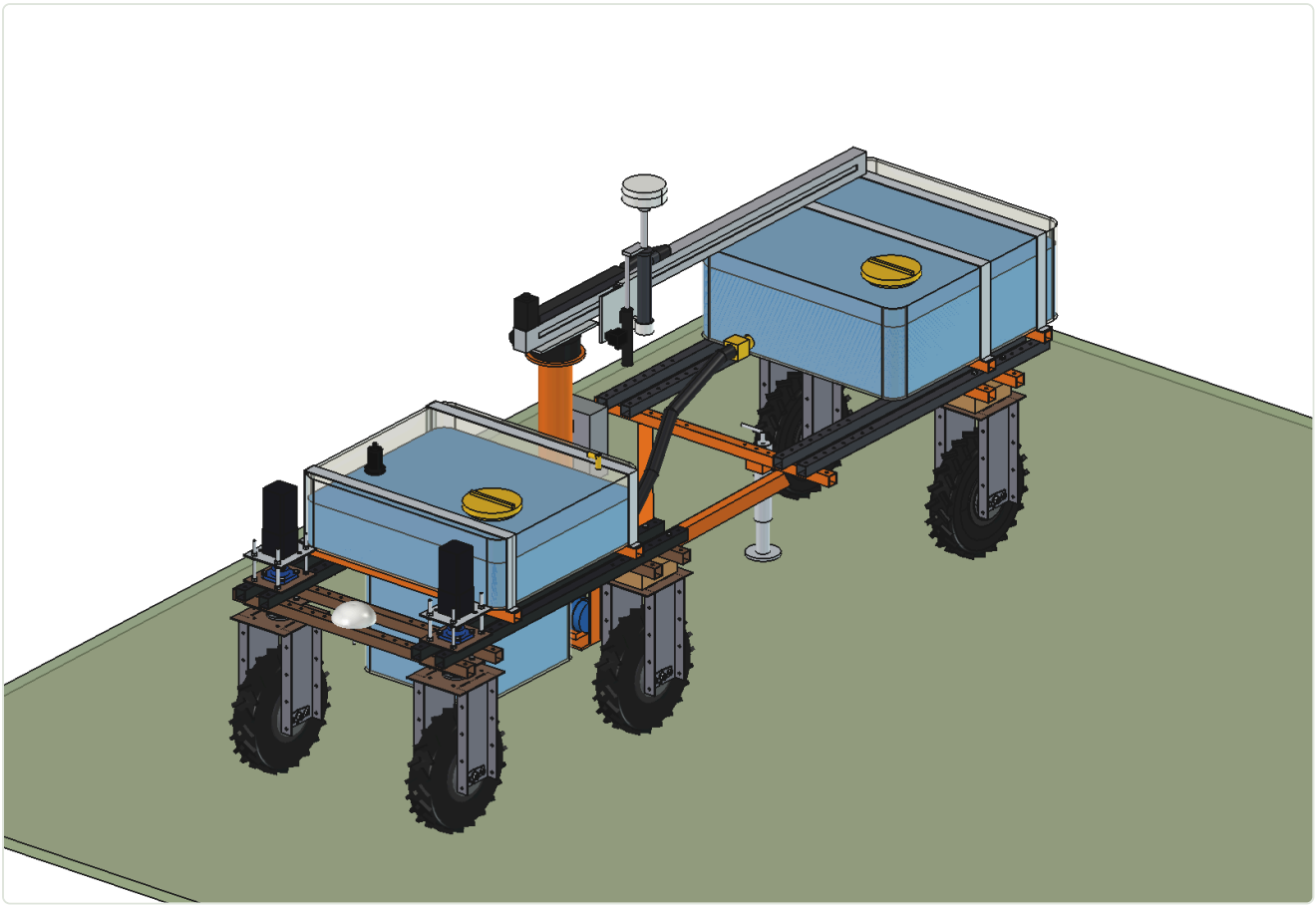
Stationary: three-point frame

5 variants: Liquid · Simple (IBC) · Stationary: IBC trailer · Stationary: flat wagon · Stationary: three-point frame

Folder: `mobile refiller/`

MOBILE REFILLER · VARIANT 1 OF 5

Liquid



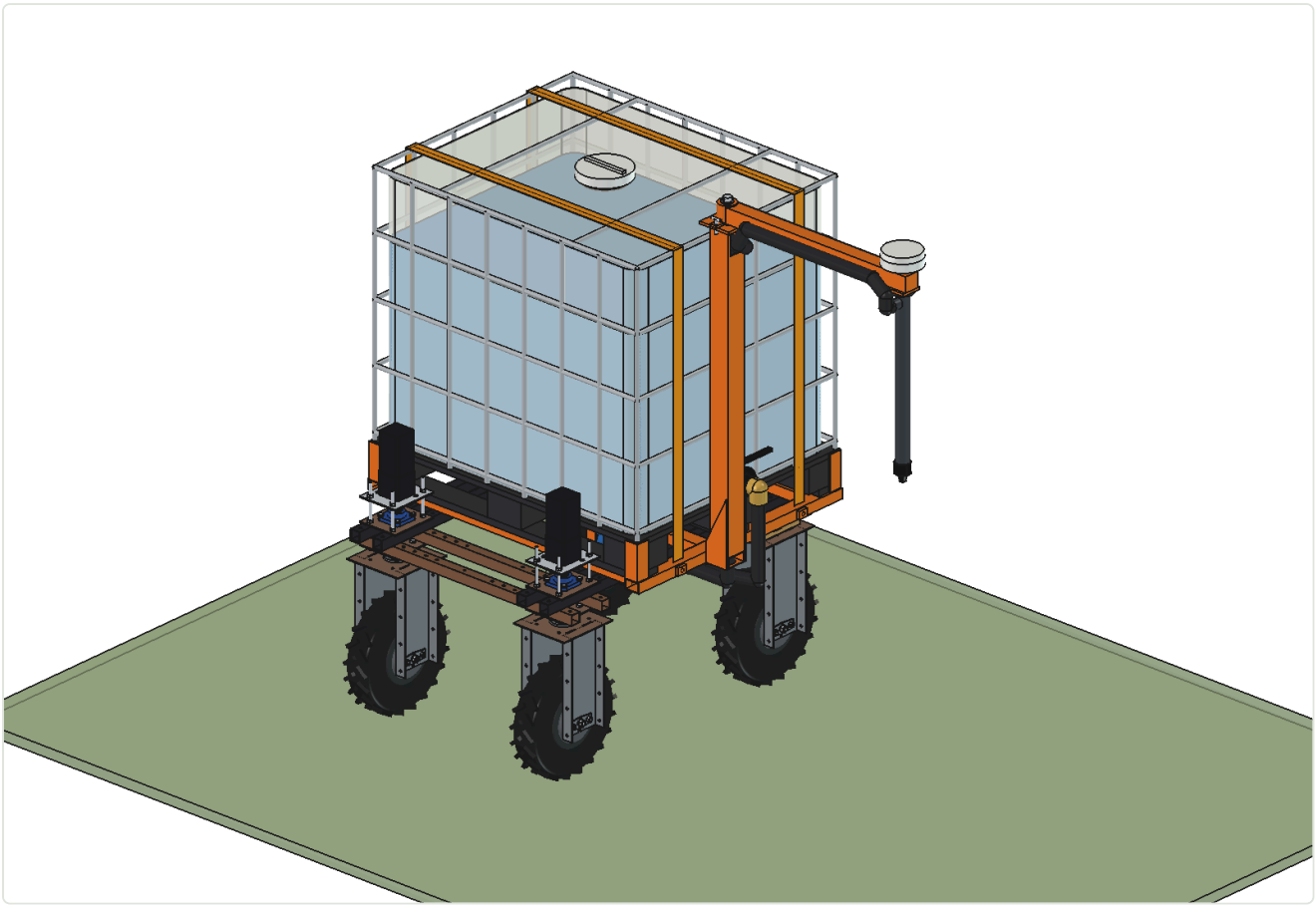
268 l T-shaped tank with a pump below the liquid level, a slewing boom with a trolley and a spout, an RTK antenna on the spout and a camera for confirmation, and a powered trailer of 246 l (6 driven wheels). Tops up the working robot in the straight reverse of its headland turn while both drive.

- About 25 l per turn, ≈ 3 s standstill
- Pump 150 l/min
- About €6,000 in parts

Folder: `mobile refiller/liquid/`

MOBILE REFILLER · VARIANT 2 OF 5

Simple (IBC)



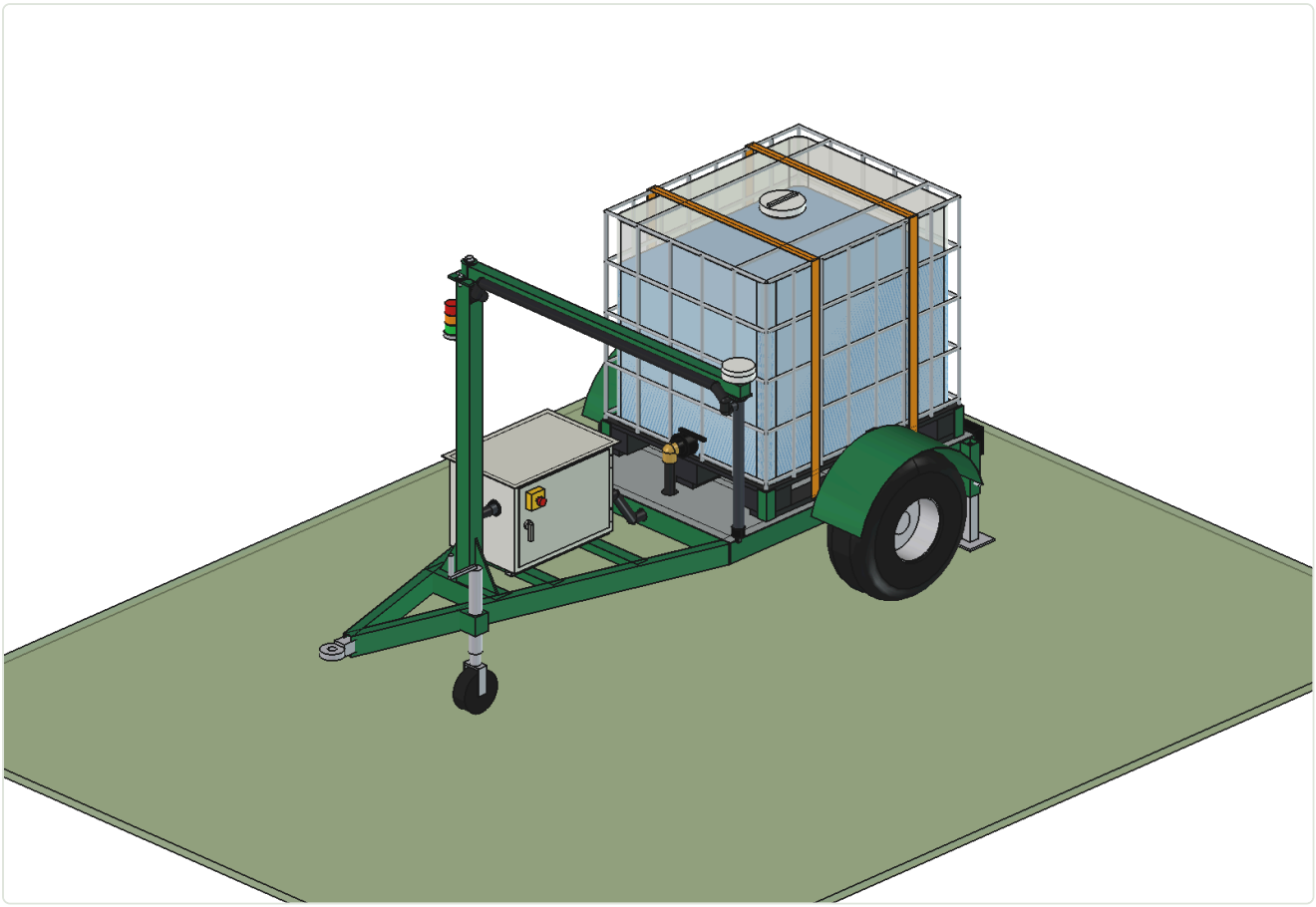
A standard 1000 l IBC set on with pallet forks, pump under the IBC, a fixed mast with a hanging hose and the RTK antenna above the nozzle. The working robot skips passes so every turn is a U-turn; the refiller stands inside the turn and the robot stops ≈ 13 s under the nozzle; 1 m further per refill.

- Standard 1000 l IBC
- About €1,650 in parts
- No boom, trailer or camera

Folder: `mobile refiller/simple/`

MOBILE REFILLER · VARIANT 3 OF 5

Stationary: IBC trailer



Single-axle trailer for one or two 1000 l IBCs, parked on the headland by a tractor. Control box (pump, battery, controller, RTK receiver, radio) on the A-frame drawbar and a mast with swinging arm, drop pipe, drip-stop nozzle and RTK antenna straight above the nozzle.

> 1 or 2 IBCs

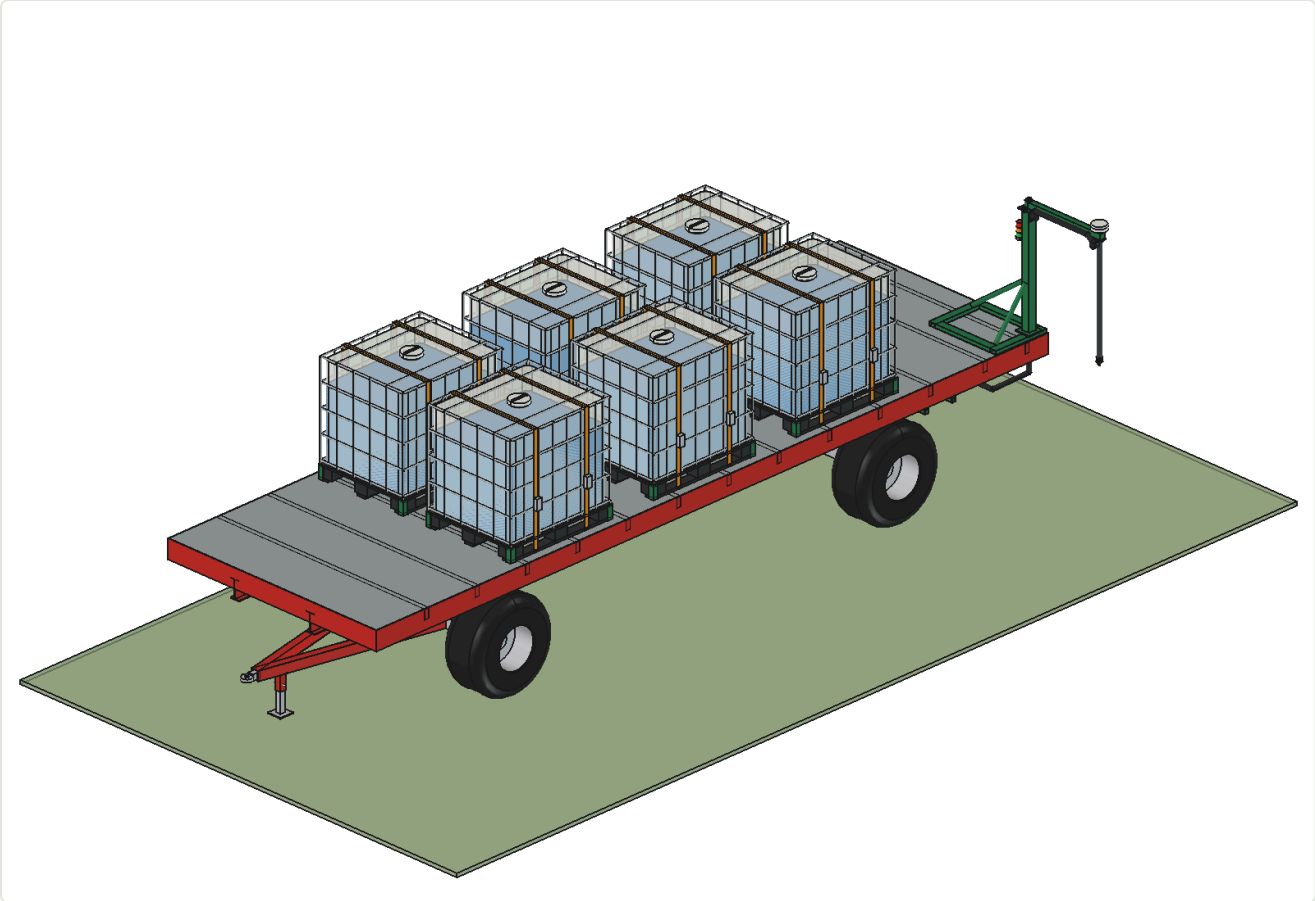
> 120 l per visit

> The working robot drives to it

Folder: `mobile refiller/stationary/ibc_trailer/`

MOBILE REFILLER · VARIANT 4 OF 5

Stationary: flat wagon



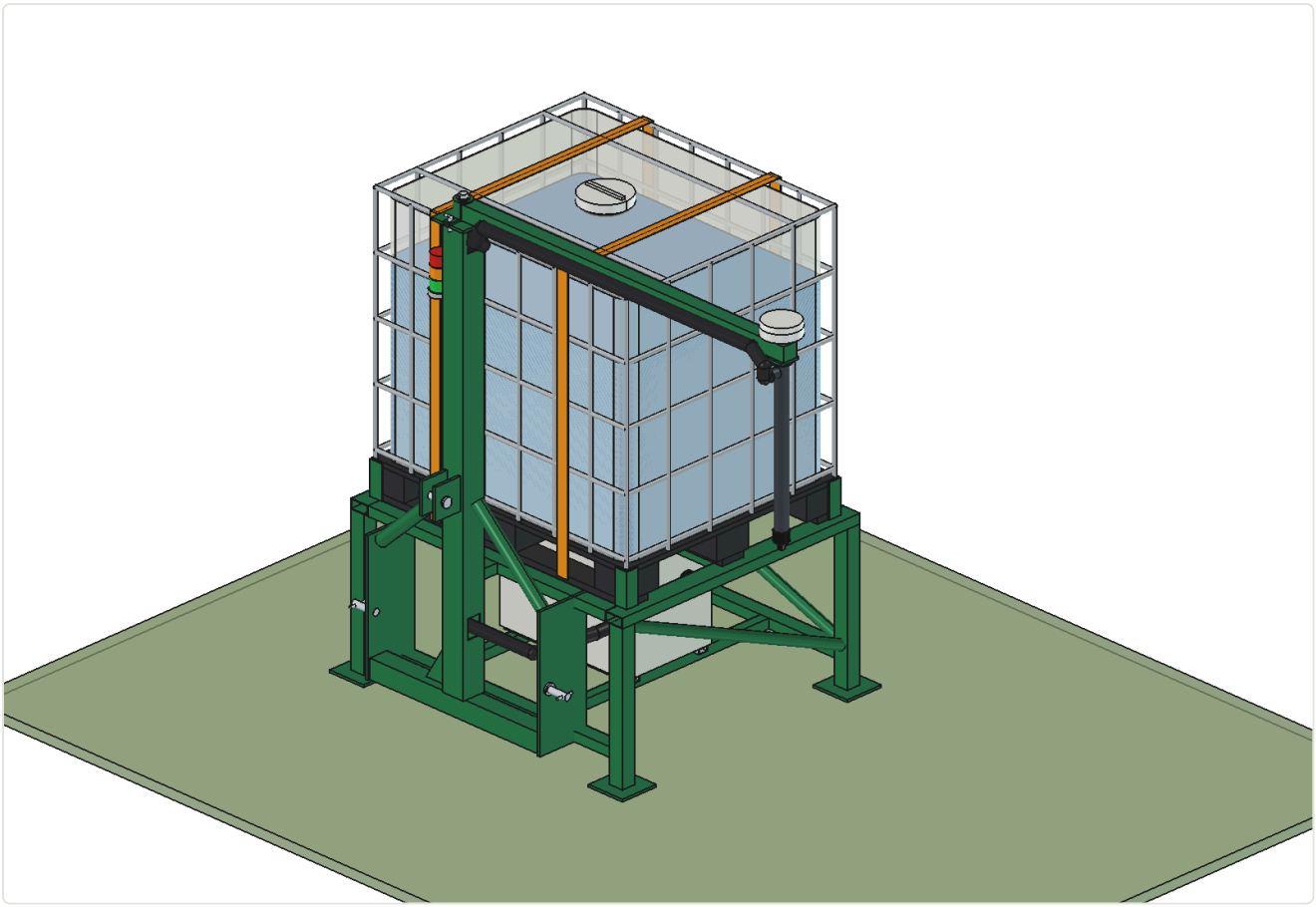
Standard two-axle flat bale wagon (8 × 2.48 m) with 6 IBCs, two across. Only two holes in the deck and two box hangers; suction header along the deck, control box under the deck, mast on a skid at the rear left corner.

- 6 IBCs
- Pump 0.8 m below the IBC bottoms
- Wagon itself unchanged

Folder: `mobile refiller/stationary/flat_wagon/`

MOBILE REFILLER · VARIANT 5 OF 5

Stationary: three-point frame



Welded frame on four legs for one 1000 l IBC, picked up on a category 2 three-point linkage and set down on the headland. The mast doubles as the hitch tower; control box under the platform.

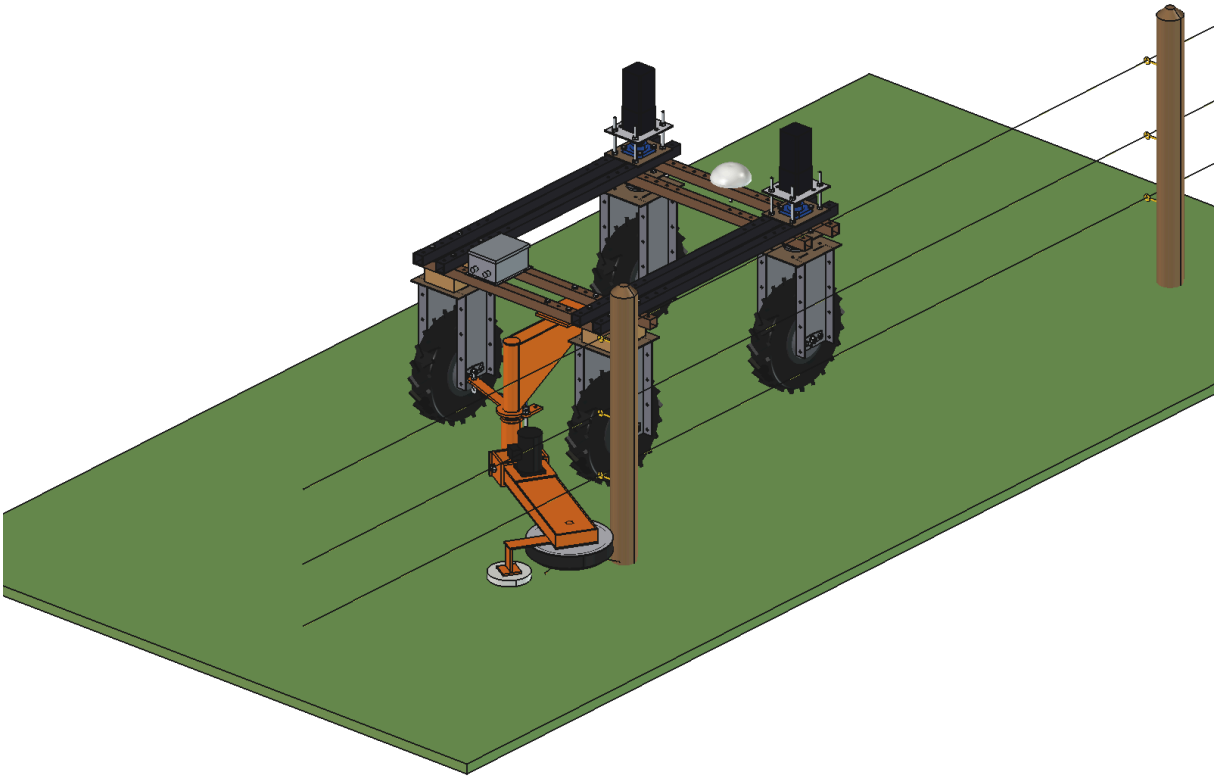
- 1 IBC
- Category 2 three-point
- Pump 535 mm below the IBC bottom

Folder: `mobile refiller/stationary/three_point/`

IMPLEMENT · MOWING

Around-pole fence mower

Mows the strip under an electric fence and goes around the posts by itself.

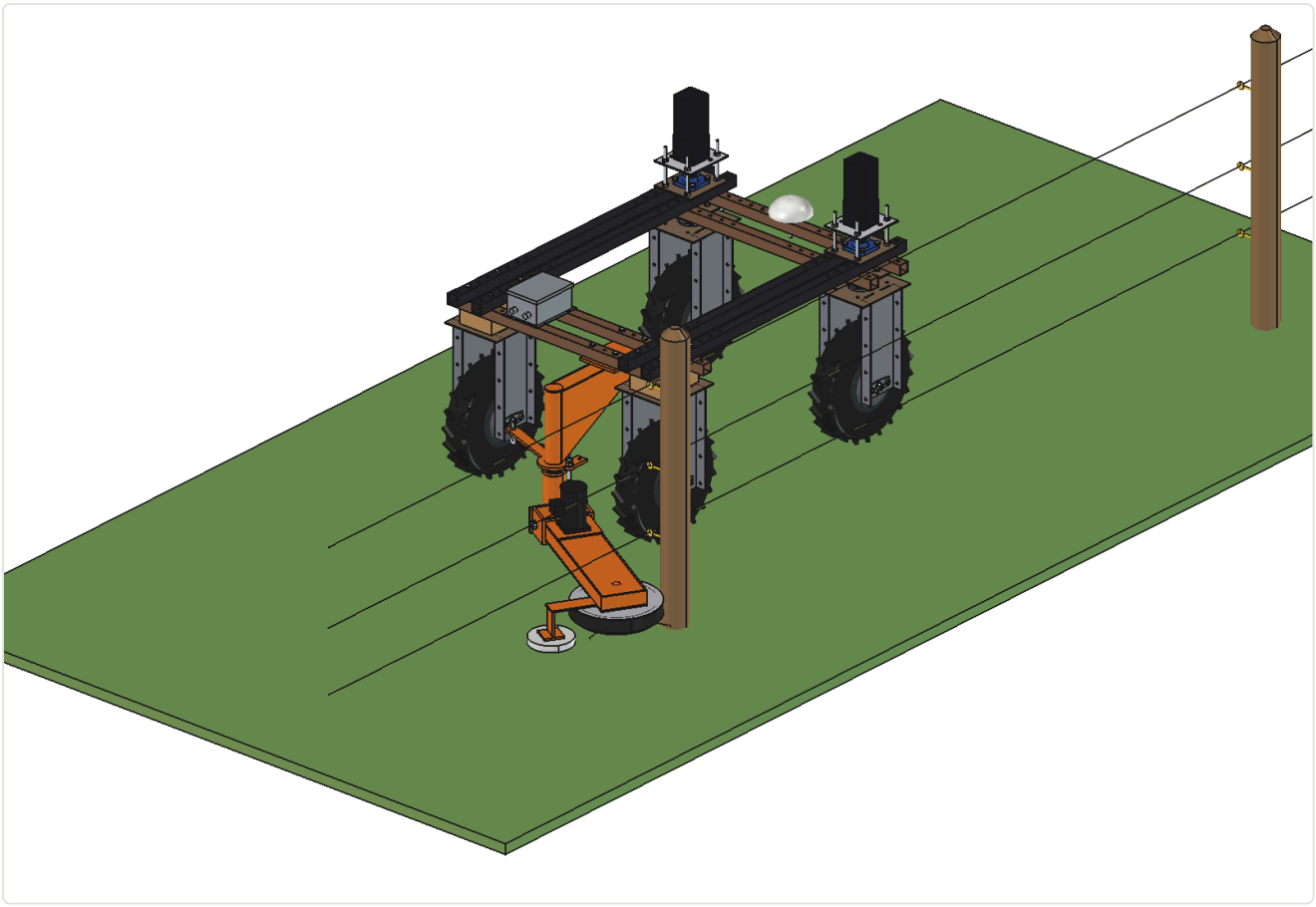


Simple

1 variant: Simple
Folder: `around_pole_mower/`

AROUND-POLE FENCE MOWER · VARIANT 1 OF 1

Simple



Trailing swing arm on a post behind the right rear wheel, held on a rest stop by a spring. A Ø400 nylon-string head under a PE-pipe guard ring slides around each post. 48 V BLDC motor at the root of the arm, V-belt inside the arm to a bought lawn-tractor deck spindle; head 182 mm high on an HDPE skid. No sensor, no actuator, no software.

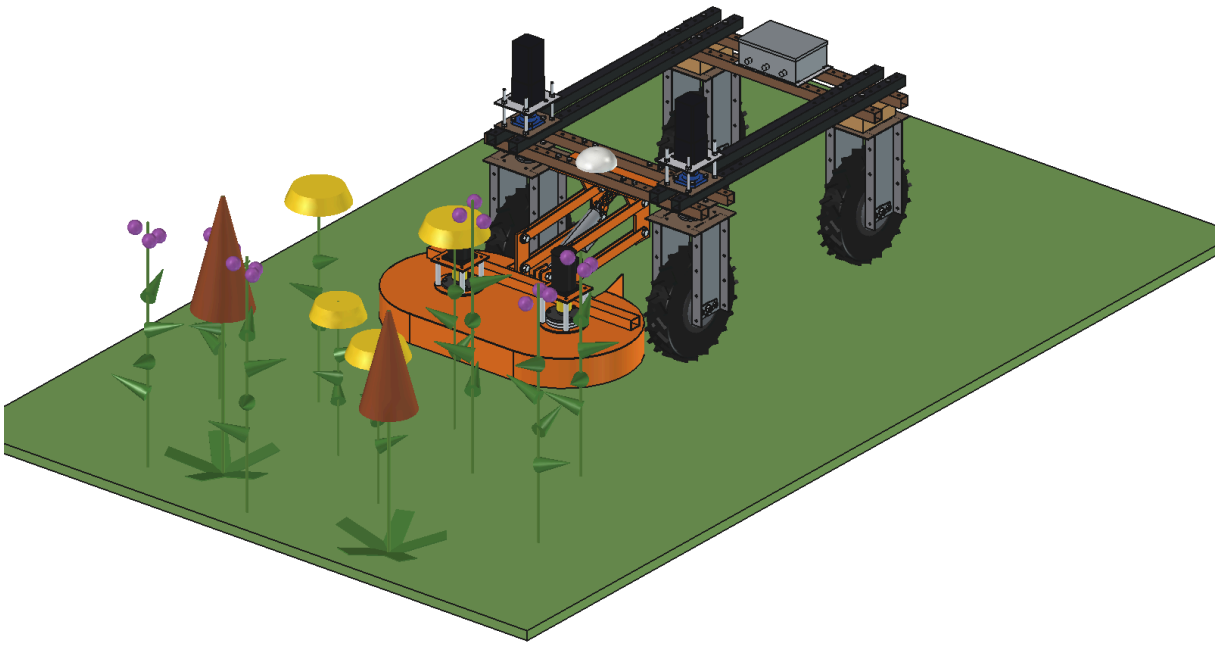
- ≈ 615 W at 0.5 m/s
- 1.8 km of fence per hour
- About €575 in parts
- Arm dynamics and animation with grass

Folder: `around_pole_mower/simple/`

IMPLEMENT · MOWING

Weed picker

Tops thistles, ragwort and dock above the grass and gathers them in one windrow, no bin. Ragwort in particular has to leave the field.

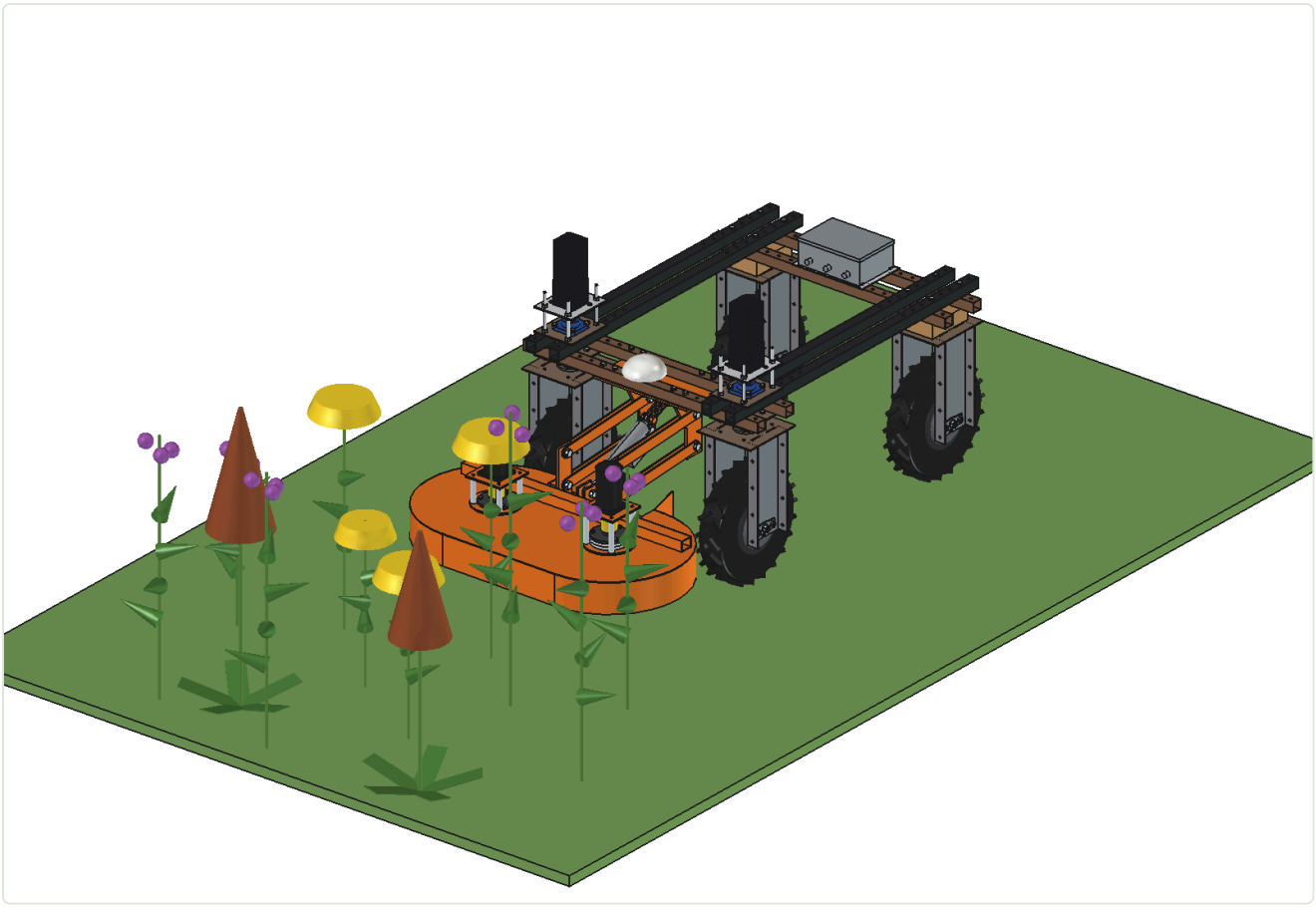


Mowing deck

1 variant: Mowing deck
Folder: `weed picker/`

WEED PICKER · VARIANT 1 OF 1

Mowing deck



Twin-rotor mowing deck in front of the robot: two rotors with free-swinging blades (Ø510, 990 mm cut) turn inwards and throw the chopped weed out of a centre opening as a windrow ≈ 320 mm wide. Each rotor: 48 V BLDC motor on a lawn-tractor deck spindle; the left rotor runs 24 mm higher. Cut height 100–400 mm from a 4000 N actuator on a parallel linkage, with a float link for humps.

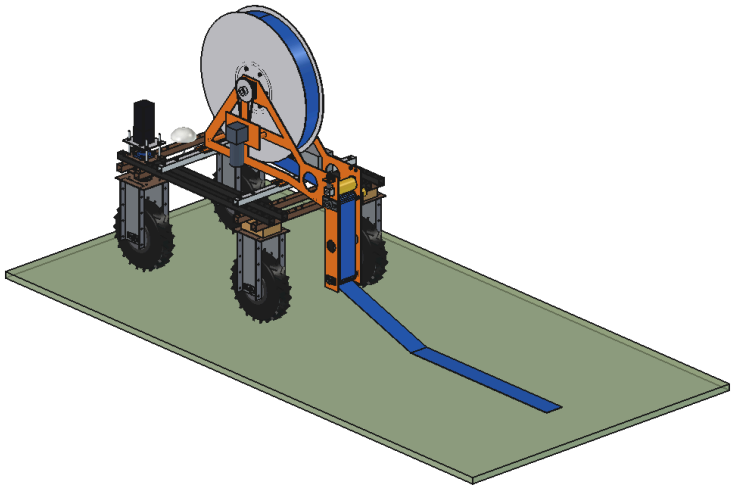
- ≈ 1 kW at 0.5 m/s
- 82 kg (rear counterweight advised)
- About €1,040 in parts
- Animation in a weed patch

Folder: `weed picker/mowing_deck/`

IMPLEMENT · SPRAYING, FERTILIZING & IRRIGATION

Irrigation

Irrigates grassland by flooding at low pressure from a pump at the ditch: little energy, no wind drift, no evaporation in the air.

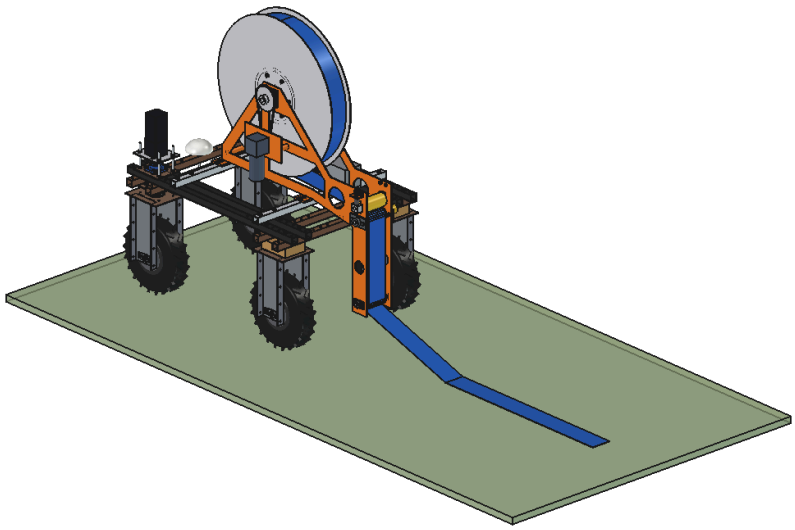


Stationary pump + hose reel

1 variant: Stationary pump + hose reel
Folder: `irrigation/`

IRRIGATION · VARIANT 1 OF 1

Stationary pump + hose reel



The robot carries a reel with 150 m of 3" lay-flat hose, lays it out empty, the petrol pump at the ditch floods a 4 m strip (outlets every 2 m), the hose runs empty and the robot winds it up again. Pinch rollers keep the reel dry and squeeze the last water out.

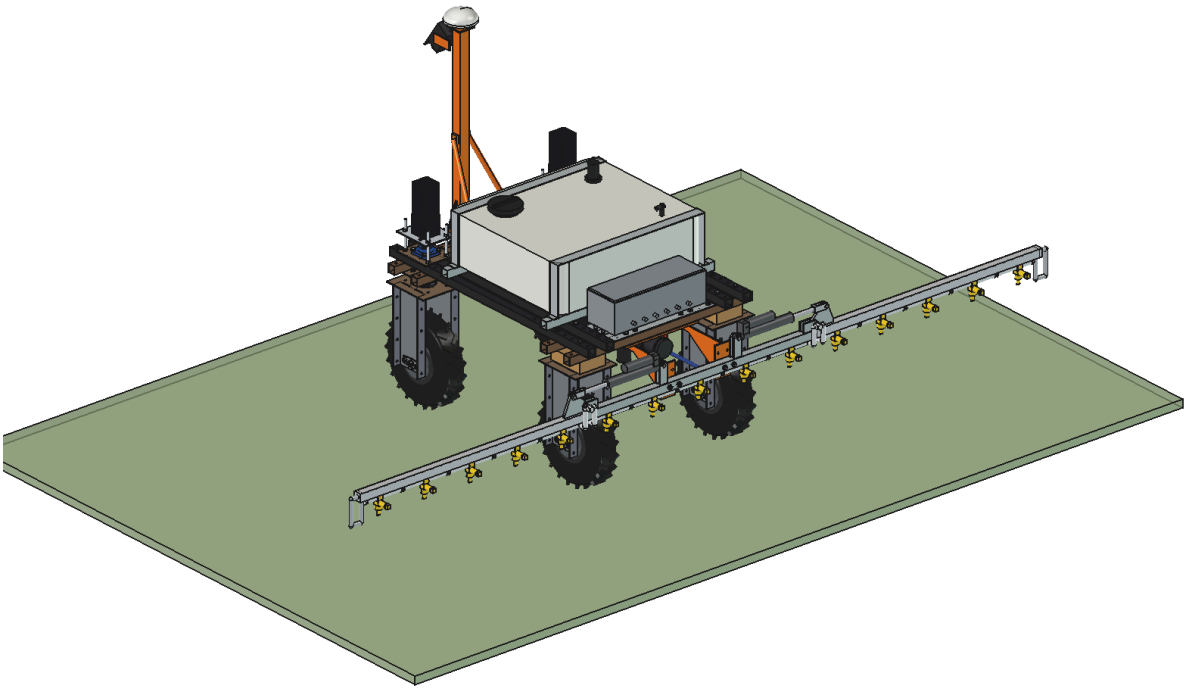
- 7.4 m^3/h at 0.17 bar, CU ≈ 99 %
- 122 min per 500 m^2 strip at 25 mm
- 146 kg, about €1,900 in parts

Folder: `irrigation/stationary + roller/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Spot sprayer

Sprays weeds plant by plant instead of the whole field. A camera in front finds the weeds; the boom behind opens only the nozzles above a plant.

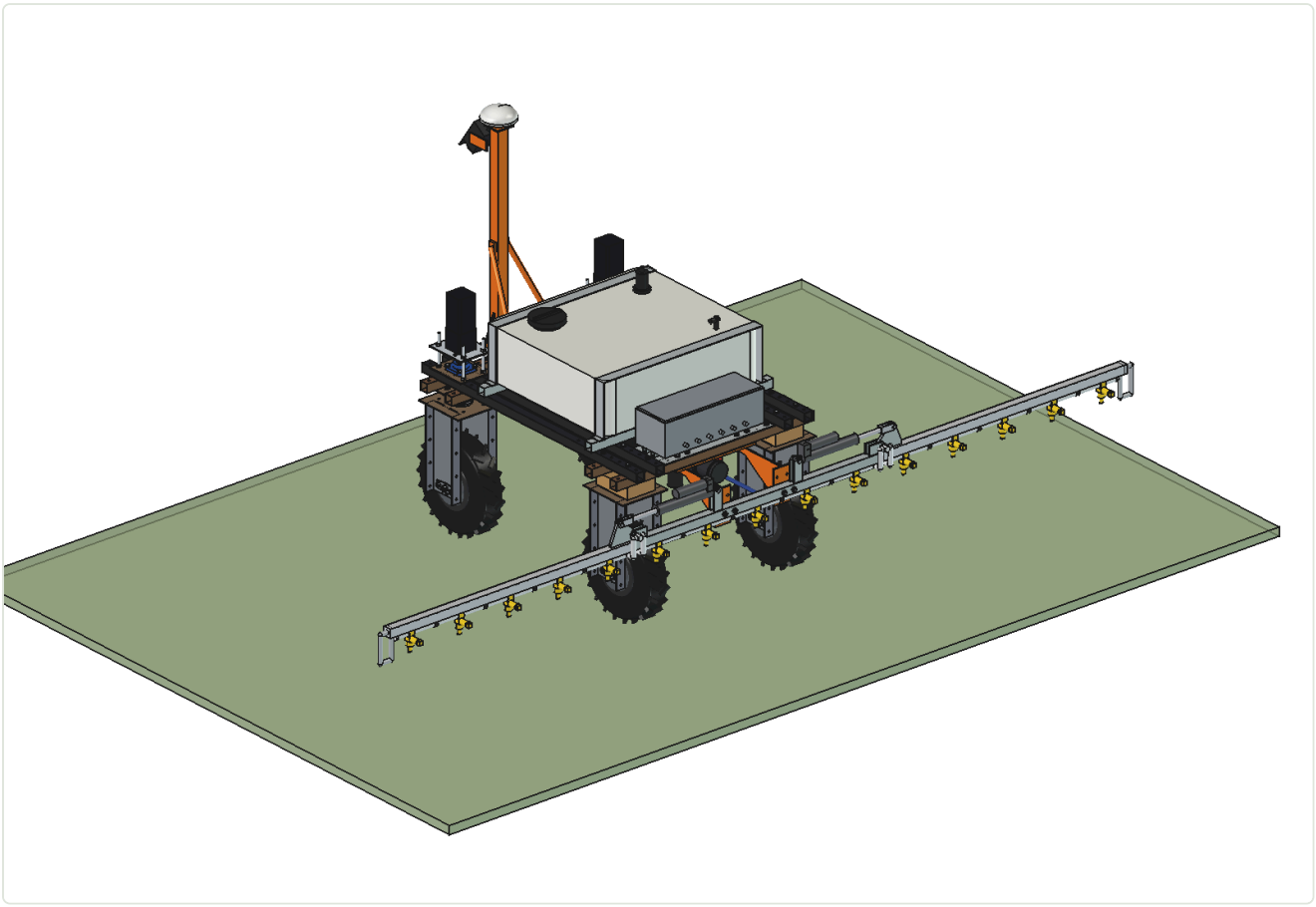


Folding 3.75 m boom

1 variant: Folding 3.75 m boom
Folder: `spotspray unit/`

SPOT SPRAYER · VARIANT 1 OF 1

Folding 3.75 m boom



Wide-angle global-shutter camera on a mast (1.48 m, 55° down) sees the whole strip from 1.8 m ahead; 2.65 s later 15 PWM nozzles at 250 mm on a stainless pipe open above the plant. Centre section 1.25 m and two 1.25 m wings that fold up in 3.2 s with a 24 V actuator each, also in the field. 120 l tank and diaphragm pump on the robot.

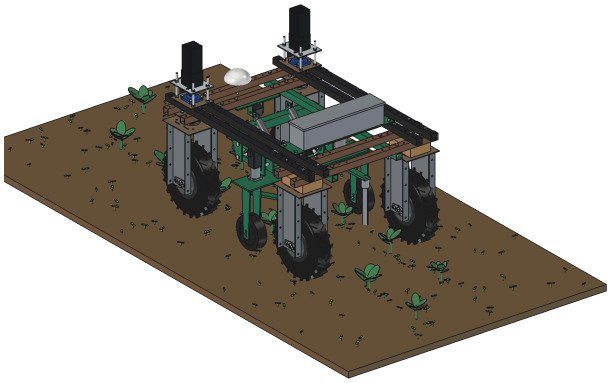
≈ 20 l/ha at 5 % weed cover, 1.1 ha/h	91 kg
About €3,580 in parts	Animation with docks and a fence post

Folder: `spotspray unit/boom/`

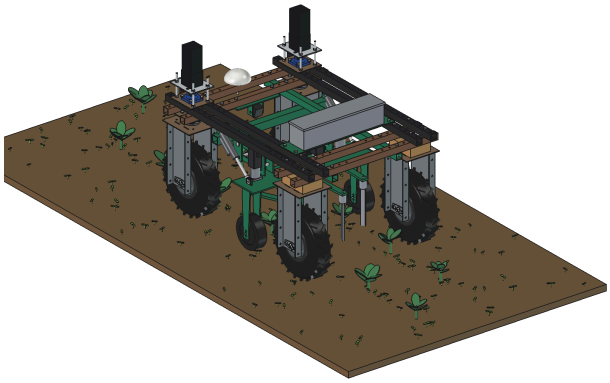
IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

In-row weeder

Hoes inside two rows of broccoli (375 mm apart, wheels in the inter-rows). Camera under the front beams finds the plants 0.84 s ahead; knives swing out around each plant.



Finger, 2 rows

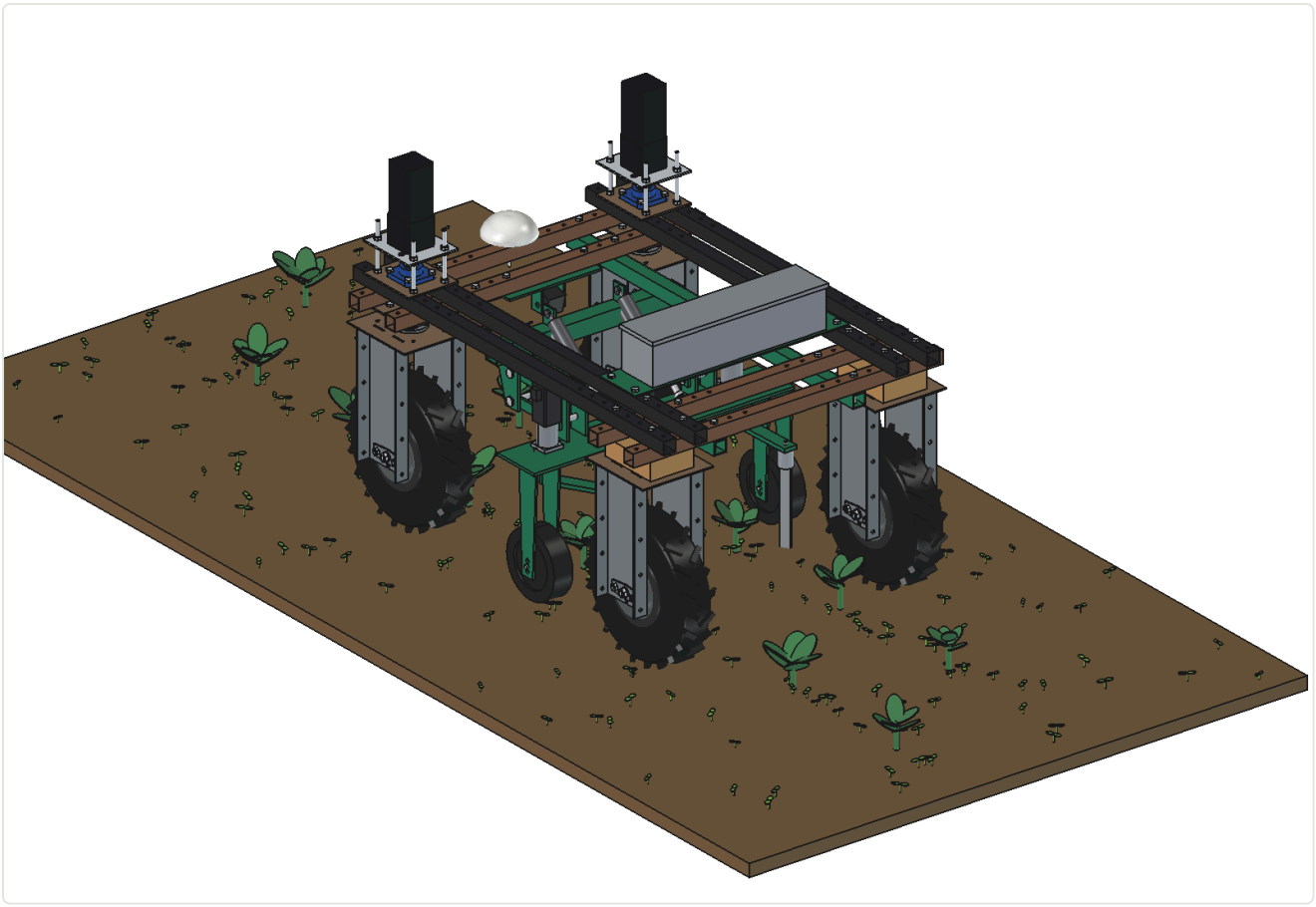


Double arms

2 variants: Finger, 2 rows · Double arms
Folder: `in row weeder/`

IN-ROW WEEDER · VARIANT 1 OF 2

Finger, 2 rows



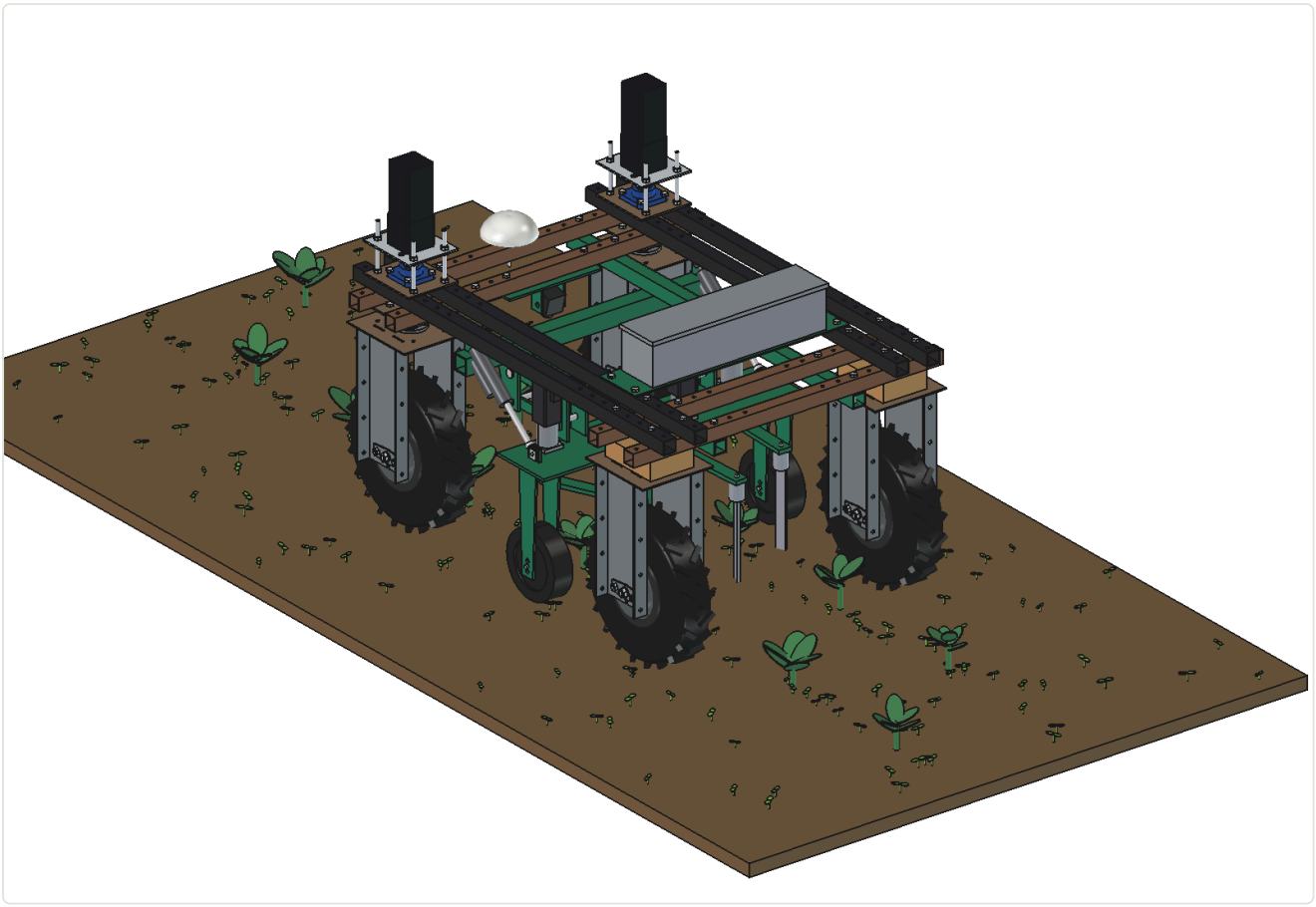
Two row units under the belly on parallelograms with gauge wheels in the robot's wheel track. A NEMA 23 closed-loop stepper with a 10:1 gearbox swings a 220 mm arm with a spring-steel finger and a flat knife 20 mm deep. Out in 98 ms, back in 148 ms; never closer than 50 mm to a stem.

- 0.5 m/s, 6,400 plants/h
- 55 kg
- About €2,240 in parts

Folder: `in row weeder/finger/`

IN-ROW WEEDER · VARIANT 2 OF 2

Double arms



Two knives per row (4 steppers) that open to both sides of each plant and close behind it, timed by the real distance to the stem. Each knife moves only 78 mm: out in 89 ms, back in 118 ms.

> 90 % of the row band outside the safe zones hoed

> 68 kg

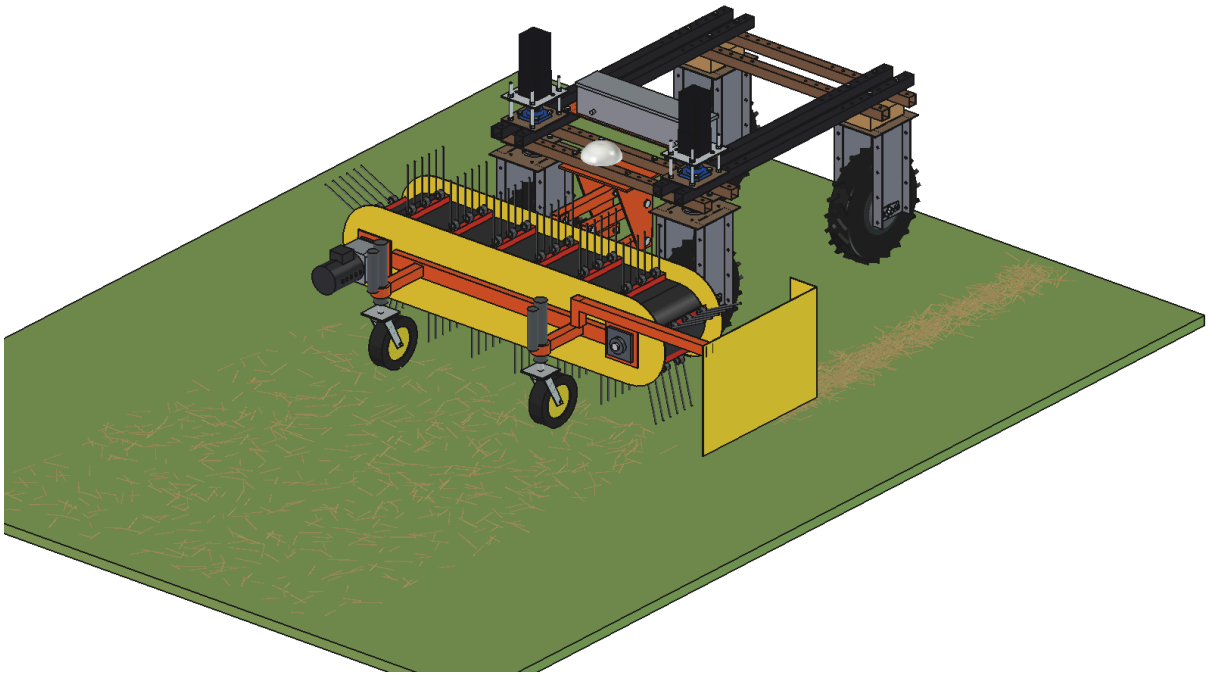
> About €2,820 in parts

Folder: `in row weeder/double_arm/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Belt rake

Band rake in front of the robot: a rubber belt with 15 tine bars around two drums (1.52 m working width). Combs weed seedlings out of nursery beds between young trees or rakes moss and thatch out of grassland into one windrow.

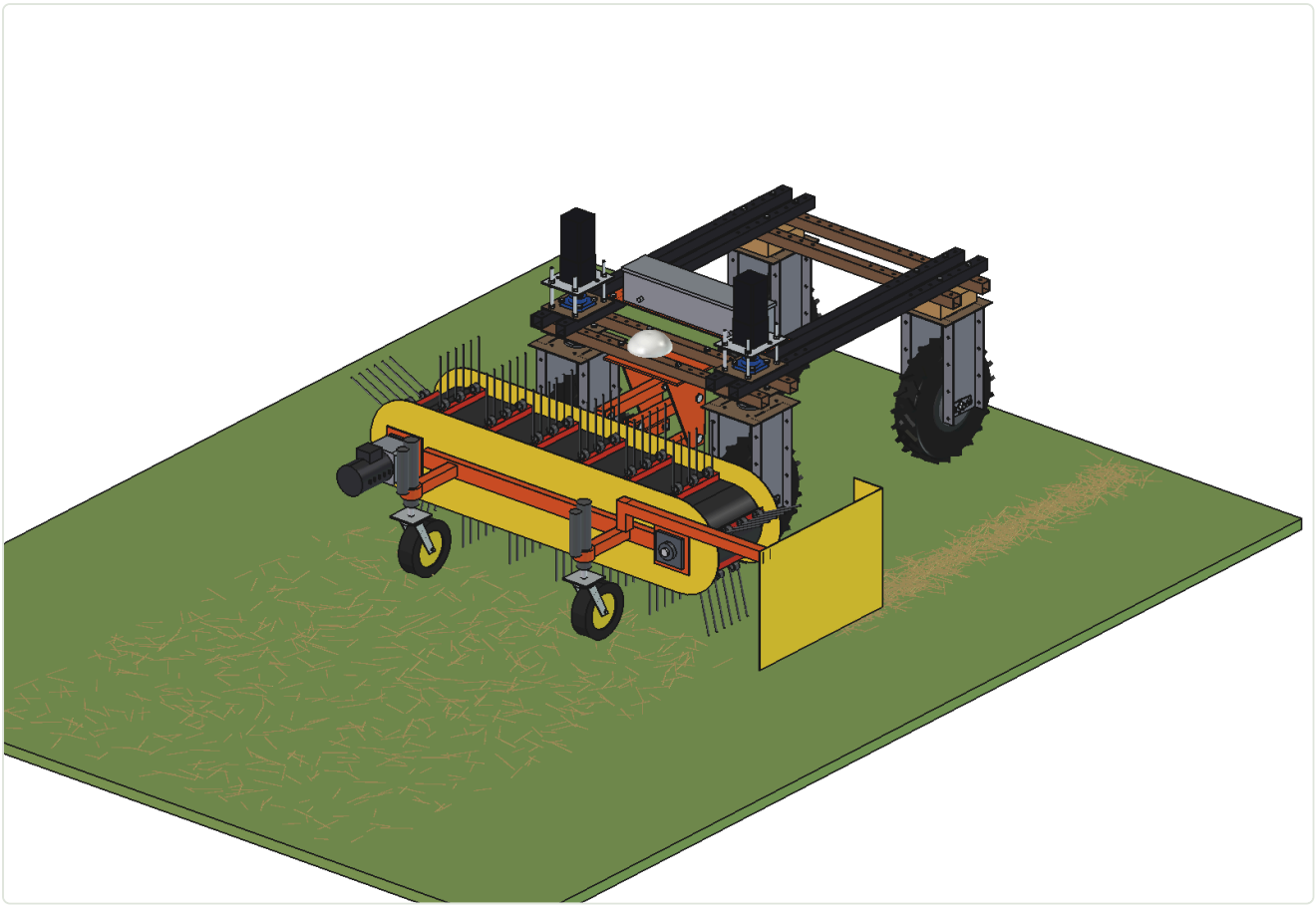


Electric

1 variant: Electric
Folder: `bandrake/`

BELT RAKE · VARIANT 1 OF 1

Electric



48 V 750 W gearmotor with a 20:1 planetary gearbox on the right drum. Two swivel castors on 24 V actuators carry the 110 kg rake, set the depth and lift it 100 mm for the headland; a free parallelogram pushes it.

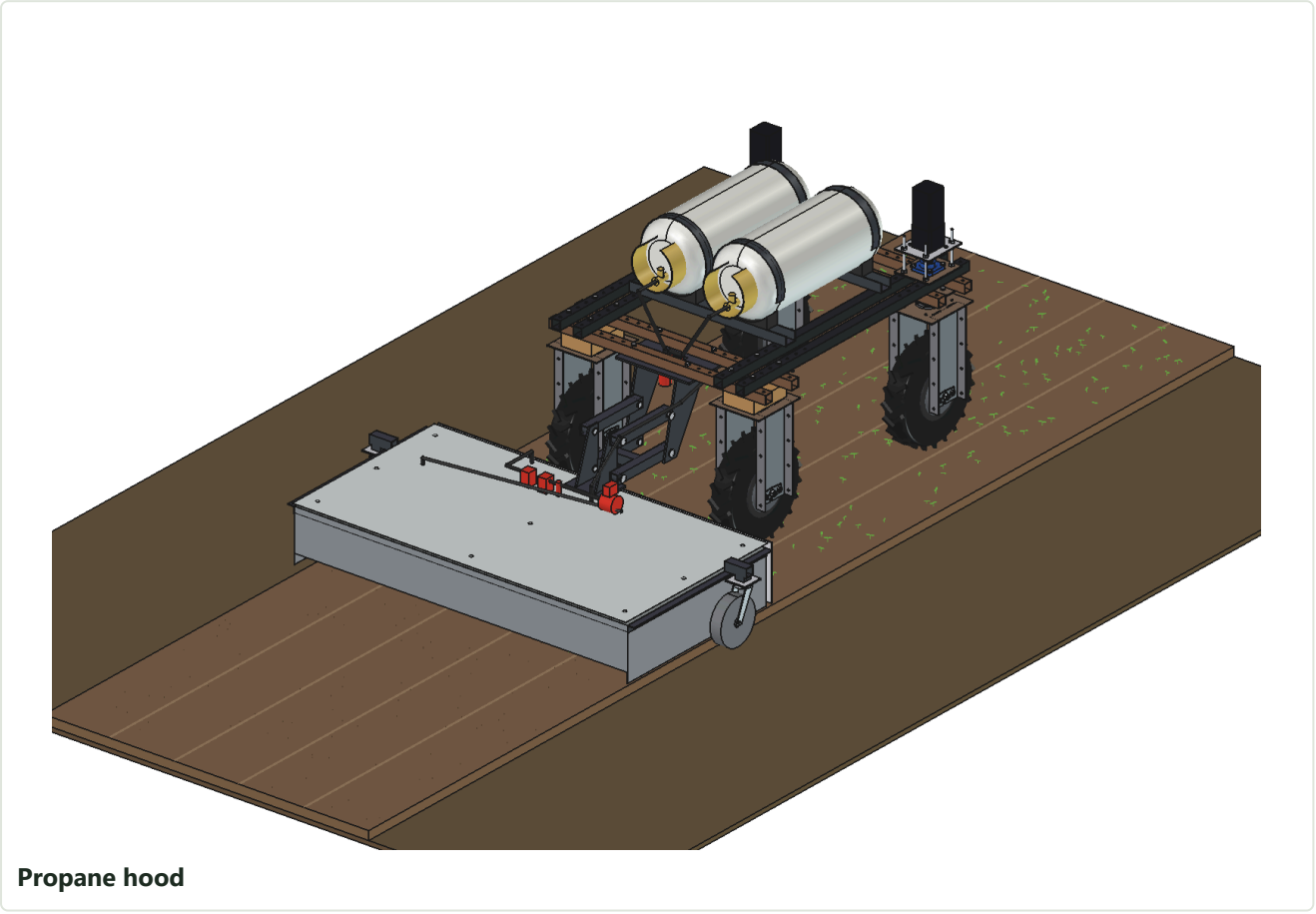
- Comb mode 0.6 m/s, rake mode 1.2 m/s, 0.49 ha/h
- About €2,380 in parts
- Tine forces and animation

Folder: `bandrake/electric/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Weeds burner (flame weeder)

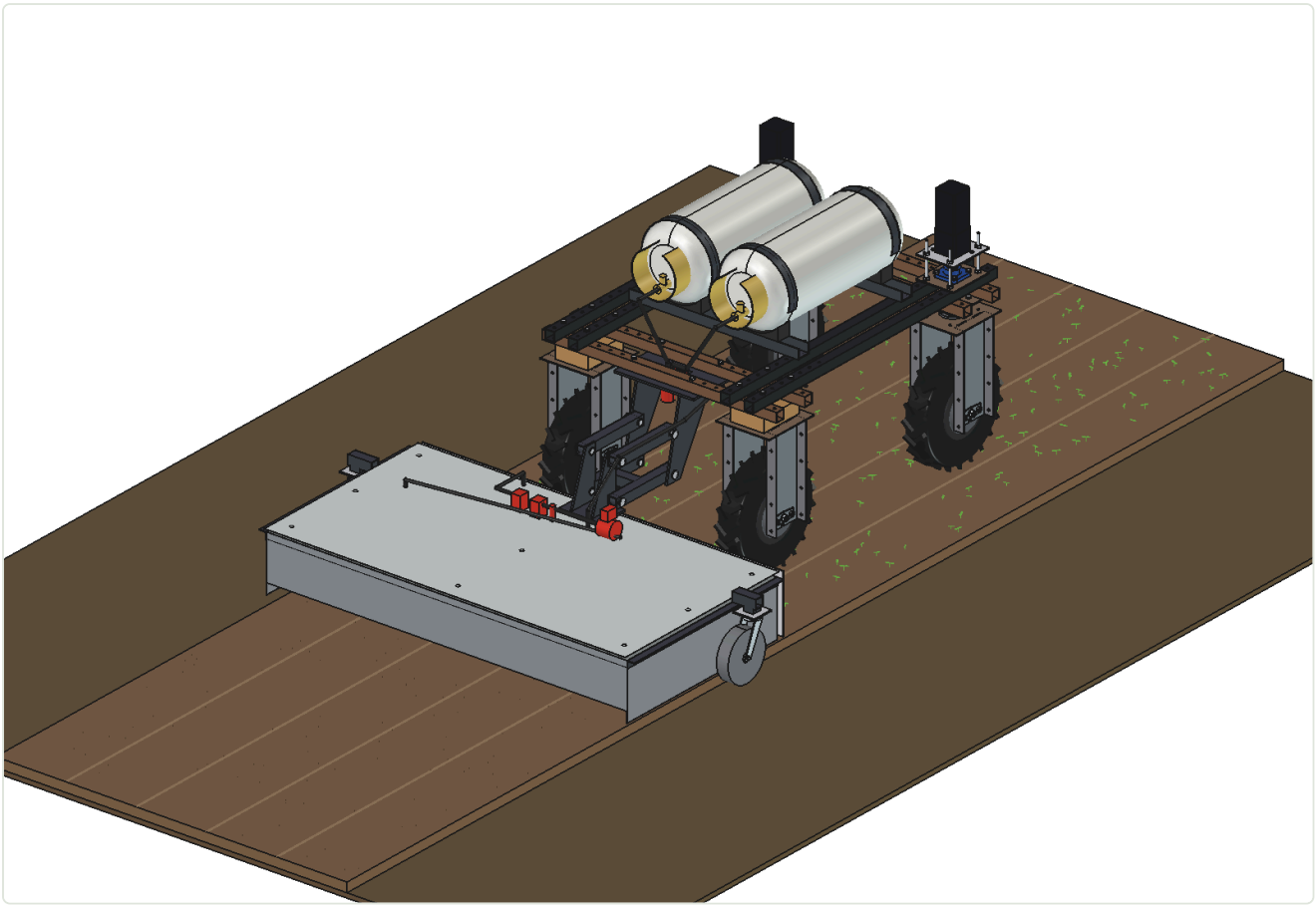
Pre-emergence flame weeder for freshly sown beds (carrot, onion, beetroot, parsnip): kills the emerged seedlings without moving the soil.



Propane hood

1 variant: Propane hood
Folder: `weeds burner/`

Propane hood



Insulated 1.5 × 0.8 m hood behind the robot with 6 flat-flame burners (13 kg/h, 167 kW). Two 18 kg forklift cylinders with liquid withdrawal on the robot, a vaporiser coil in the hood, two main solenoid valves in series, a motorised regulator, pilots with igniters and flame sensors; the valves close at once when the robot stops.

- 34 kg/ha at 0.7 m/s, 0.84 ha per filling
- Gas ≈ €55/ha
- 131 kg empty, about €3,400 in parts

Folder: `weeds burner/propane_hood/`

IMPLEMENT IDEAS

Ideas not worked out yet

31 attachments from the website's list that have no model yet. Each has a folder in `implement ideas/` with a `README.md` describing what it does, so it can be worked out later.

Soil Cultivation & Weed Control

- **(Rubber) push blade / grader blade**
- **Small cultivator**
- **Single-axis inter-row hoe**
- **Drop-on-demand boom (coarse droplets)**
- **Grassland aerator**

Mowing

- **Sickle bar mower**

Spraying, Fertilizing & Irrigation

- **Irrigation pump (direct pump)**
- **Fertilizer spreader**
- **Lime spreader**

Sowing & Planting

- **Tray setter**
- **Patch seeder**

Spreading & Bedding

- **Sawdust spreader**
- **Straw spreader**
- **Compost spreader / bin**

Power & Drive

- **Battery pack**
- **Generator / engine**
- **Solar panels**
- **Heavy-duty electric motors**

Transport & General

- **Broom / sweeper**
- **Seat & frame extension**
- **Dumper / power barrow**
- **Tow hitch**
- **Counterweights**
- **Hydraulic pump**
- **Three-point hitch**
- **Small cabin**

Measuring & Sensors

- **Grass meter / crop scanner**
- **Soil compaction meter**
- **NIR sensor (NDVI & crop status)**

Cooperation & Swarm Operation

- **Drone combination**
- **Swarm operation (multiple robots per field)**

IDEA · SOIL CULTIVATION & WEED CONTROL

(Rubber) push blade / grader blade

Not worked out yet

WHAT IT DOES

Front-mounted or mid-mounted between the front and rear wheels. Usable as a grader for levelling terrain, for pushing feed and grass back in the stable, or for cleaning silos.

CONCEPT (FIRST THOUGHTS)

A wide blade with a replaceable rubber or steel wearing edge on a pivoting frame, lifted by a linear actuator like the lift frames of the other implements. A hinge with a float stop lets the blade follow the floor; an angled (dozer) setting moves material to one side. Front-mounted it pushes, mid-mounted it scrapes under the belly.

REUSE FROM EXISTING IMPLEMENTS

Lift frame and headstock of the liquid fertilizer applicator or overseeder; the push-back job overlaps with the feed pusher auger.

OPEN QUESTIONS

- Blade width: robot width (850 mm) or wider with side wings?
- Push force against wheel grip: how much load can the 4 wheels transmit?
- Rubber edge for concrete versus steel edge for soil
- Is the blade angle adjustable, and by what (actuator or pins)?

Folder: `implement ideas/push_blade/`

IDEA · SOIL CULTIVATION & WEED CONTROL

Small cultivator

Not worked out yet

WHAT IT DOES

Compact cultivator for soil cultivation in vegetable gardens and light plots.

CONCEPT (FIRST THOUGHTS)

A toolbar with 3 to 5 spring or rigid tines on a parallel linkage, depth set by gauge wheels. Light soil only, so draft force stays within what the robot can pull. Optionally a rolling basket behind the tines.

REUSE FROM EXISTING IMPLEMENTS

Toolbar, parallelogram and gauge wheels of the in-row weeder and the liquid fertilizer applicator.

OPEN QUESTIONS

- Draft force per tine in light soil (measure)
- Tine spacing versus working width
- Ballast needed to keep the tines in the ground

Folder: `implement ideas/small_cultivator/`

IDEA · SOIL CULTIVATION & WEED CONTROL

Single-axis inter-row hoe

Not worked out yet

WHAT IT DOES

Hoeing unit with single-axis (left/right) guidance that stays tightly between the rows, for example in maize, so weeds are cut close to the crop without damaging it.

CONCEPT (FIRST THOUGHTS)

A hoe frame with A-blades or goosefoot shares that is shifted sideways by a linear actuator or stepper. A camera or the RTK row map gives the lateral error; the shift axis corrects it. Gauge wheels in the robot's wheel track keep the depth.

REUSE FROM EXISTING IMPLEMENTS

Parallelogram unit with gauge wheel and camera of the in-row weeder; the X stage idea of the field marker.

OPEN QUESTIONS

- Lateral stroke and speed of the shift axis
- Row crop (maize, beet) and row distance to design for
- Camera-guided versus RTK-guided
- Protection against crop damage when the guidance drops out

Folder: `implement ideas/inter_row_hoe/`

IDEA · SOIL CULTIVATION & WEED CONTROL

Drop-on-demand boom (coarse droplets)

Not worked out yet

WHAT IT DOES

Boom with individually controlled injectors that fire single coarse droplets on demand at low pressure. Every droplet is placed straight onto the weed, which practically eliminates drift and cuts chemical usage to a fraction of a conventional spray pass.

CONCEPT (FIRST THOUGHTS)

Same carrier as the spot sprayer (camera mast in front, boom behind), but with drop-on-demand valves instead of PWM flat-fan nozzles. Droplets are fired at the detected plant position, so the boom needs a finer position resolution along the boom than 250 mm.

REUSE FROM EXISTING IMPLEMENTS

Camera mast, tank, pump, folding boom and control box of the spot sprayer (spotspray unit/boom).

OPEN QUESTIONS

- Injector type, droplet size and firing rate
- Injector pitch along the boom
- Aiming accuracy at 0.5 m/s: time from camera to droplet
- Product types and viscosity the injectors can handle

Folder: `implement ideas/drop_on_demand_boom/`

IDEA · SOIL CULTIVATION & WEED CONTROL

Grassland aerator

Not worked out yet

WHAT IT DOES

Improving soil structure and air supply in grassland.

CONCEPT (FIRST THOUGHTS)

Spiked or hollow-tine rollers, or a row of spring-loaded slitting knives, behind or in front of the robot. The element presses down with robot weight and ballast; a rolling drum carries the tines so no extra drive is needed.

REUSE FROM EXISTING IMPLEMENTS

Floating lift frame with gas springs of the overseeder (the aerator can be combined with it as a slot cutter); lifted by the same actuator.

OPEN QUESTIONS

- Tine type: solid spike, hollow tine or slitting knife
- Down force needed for penetration depth
- Passive roller versus driven roller
- Combining with the overseeder in one pass

Folder: `implement ideas/grassland_aerator/`

IDEA · MOWING

Sickle bar mower

Not worked out yet

WHAT IT DOES

Mowing at various heights, including 'topping' plants above ground level.

CONCEPT (FIRST THOUGHTS)

A bought double-knife or single-knife cutter bar on a parallel linkage in front of the robot, with a 48 V motor and a crank drive. Height by an actuator, a float link lets it follow the ground. The weed picker already tops weeds with rotors; the bar gives a clean cut with low power.

REUSE FROM EXISTING IMPLEMENTS

Parallel linkage, actuator, float link and 48 V drive of the weed picker (weed picker/mowing_deck).

OPEN QUESTIONS

- Bar length (robot width or wider)
- Power per metre in grass and in thick weeds
- Cut-material handling: left in swath or gathered
- Stone and molehill protection (breakaway)

Folder: `implement ideas/sickle_bar_mower/`

IDEA · SPRAYING, FERTILIZING & IRRIGATION

Irrigation pump (direct pump)

Not worked out yet

WHAT IT DOES

Pumping water from ditches for local irrigation or crop spraying (website: Irrigation Pump). In the repo this is the not-yet-worked-out variant "direct pump" of the irrigation implement: pump on the robot, water from a tank, a ditch or a hydrant.

CONCEPT (FIRST THOUGHTS)

Instead of a petrol pump staying at the ditch, a pump on the robot feeds a hose or a short boom directly. Water comes from a tank on a trailer or a hydrant at the field edge.

REUSE FROM EXISTING IMPLEMENTS

Reel, pinch rollers and outlets of irrigation/stationary + roller; the mobile refiller for tank supply.

OPEN QUESTIONS

- Pump mass on the robot versus pump at the ditch
- Hose or boom as distributor
- Water supply: tank, hydrant or ditch
- Comparison of energy and capacity with the stationary-pump variant

Folder: `implement ideas/irrigation_direct_pump/`

IDEA · SPRAYING, FERTILIZING & IRRIGATION

Fertilizer spreader

Not worked out yet

WHAT IT DOES

Precise application and spreading of granular fertilizers.

CONCEPT (FIRST THOUGHTS)

A hopper with a metering roller and a spreading disc or a drop tube per row. Dose follows the driving speed (as with the overseeder's cam rollers) and can follow a task map from the crop scanner for variable-rate application.

REUSE FROM EXISTING IMPLEMENTS

Metering rollers and speed-following motors of the overseeder; granular loading from the mobile refiller concept.

OPEN QUESTIONS

- Spreading disc versus row drop tubes
- Working width and edge spreading accuracy
- Hopper size versus robot axle load
- Variable-rate from a task map: update rate

Folder: `implement ideas/fertilizer_spreader/`

IDEA · SPRAYING, FERTILIZING & IRRIGATION

Lime spreader

Not worked out yet

WHAT IT DOES

Spreading lime to improve soil pH levels and structure.

CONCEPT (FIRST THOUGHTS)

Like the fertilizer spreader, but with a coarser metering roller or belt for powdery or granular lime and a covered outlet against dust. Large dose per hectare means large hopper volume, so reloading logistics matter.

REUSE FROM EXISTING IMPLEMENTS

Fertilizer spreader hopper and metering; mobile refiller for reloading.

OPEN QUESTIONS

- Dose per hectare and hopper volume
- Dust control and flow behaviour of damp lime
- Reloading: how often per hectare?

Folder: `implement ideas/lime_spreader/`

IDEA · SOWING & PLANTING

Tray setter

Not worked out yet

WHAT IT DOES

Front-mounted unit that takes seedlings out of the tray and sets them out at the right spacing.

CONCEPT (FIRST THOUGHTS)

A magazine of trays above the front beams, a gripper or pin-picker takes one seedling per cycle and plants it in a dibbled hole. Spacing by the robot's odometry; the robot stops or crawls per plant.

REUSE FROM EXISTING IMPLEMENTS

XYZ stage idea of the field marker for positioning; camera and odometry of the in-row weeder.

OPEN QUESTIONS

- Tray type and plug size
- Pick-and-place mechanism and cycle time
- Planting depth and soil closing
- Tray reloading, again from a refiller?

Folder: `implement ideas/tray_setter/`

IDEA · SOWING & PLANTING

Patch seeder

Not worked out yet

WHAT IT DOES

Compact seeder that spreads grass seed onto bare patches in the sward, so damaged spots close up again quickly.

CONCEPT (FIRST THOUGHTS)

A small hopper with a short spreader over a scratching or raking element, switched on only above bare patches that a camera finds (as the spot sprayer finds docks). Seed is dosed per patch, not per hectare.

REUSE FROM EXISTING IMPLEMENTS

Camera detection of the spot sprayer; seed hopper and metering of the overseeder; rake tines of the belt rake.

OPEN QUESTIONS

- Patch detection: bare soil in grass
- Seed dose per patch and spreading width
- Seed-to-soil contact: rake, roll or press
- Combined with the overseeder or separate?

Folder: `implement ideas/patch_seeder/`

IDEA · SPREADING & BEDDING

Sawdust spreader

Not worked out yet

WHAT IT DOES

Even distribution of sawdust or bedding material in stables.

CONCEPT (FIRST THOUGHTS)

A bin on the robot with a bottom belt or screw and a throwing wheel or distributor, so a stable can be bedded without a tractor. Hopper height must clear the stable doors; weight is balanced over the axles.

REUSE FROM EXISTING IMPLEMENTS

Auger and hood design of the feed pusher auger; mobile refiller for reloading.

OPEN QUESTIONS

- Bin size versus door height and axle load
- Throwing width and evenness
- Dust and moisture of the material
- Stable floor and slope

Folder: `implement ideas/sawdust_spreader/`

IDEA · SPREADING & BEDDING

Straw spreader

Not worked out yet

WHAT IT DOES

Evenly spreading straw, specially designed for use between vegetable rows.

CONCEPT (FIRST THOUGHTS)

A bale rack or bin with a shredding and metering unit that drops straw as a narrow strip between rows, for weed suppression and moisture. Width limited to the inter-row space of 375 mm rows (the wheels run in the inter-rows).

REUSE FROM EXISTING IMPLEMENTS

Row geometry of the in-row weeder; metering from the sawdust spreader.

OPEN QUESTIONS

- Bale handling or loose straw bin
- Strip width and thickness between rows
- Blocking of the metering unit by long straw

Folder: `implement ideas/straw_spreader/`

IDEA · SPREADING & BEDDING

Compost spreader / bin

Not worked out yet

WHAT IT DOES

Transporting compost within the frame structure and spreading it between the crop rows.

CONCEPT (FIRST THOUGHTS)

A bin between the beams with a bottom belt or screw that meters compost to drop tubes between the rows. Heavy payload, so axle load and the low centre of gravity matter more than for the other spreaders.

REUSE FROM EXISTING IMPLEMENTS

Metering of the straw and sawdust spreaders; axle-load calculation of the *_calc.py files.

OPEN QUESTIONS

- Payload versus wheel and motor ratings
- Bin shape for flow of damp compost
- Metering accuracy and bridging of the material

Folder: `implement ideas/compost_spreader/`

IDEA · POWER & DRIVE

Battery pack

Not worked out yet

WHAT IT DOES

Additional power supply modules to be installed between the frames.

CONCEPT (FIRST THOUGHTS)

A removable battery module in a protective case, bolted between the beams on the 50 mm grid, connected to the 48 V bus through a fuse and contactor. Fits as ballast too, so it can double as a counterweight.

REUSE FROM EXISTING IMPLEMENTS

Control-box mounting of the mobile refiller and the weeds burner.

OPEN QUESTIONS

- Cell format, capacity and mass
- Connector and fuse rating
- Battery management and charging
- Quick-swap mechanism

Folder: `implement ideas/battery_pack/`

IDEA · POWER & DRIVE

Generator / engine

Not worked out yet

WHAT IT DOES

Generator or combustion engine mounted between the frame as an auxiliary power source for extra electricity or mechanical power.

CONCEPT (FIRST THOUGHTS)

A small generator (range extender) on a vibration-damped frame between the beams, charging the 48 V bus. Exhaust and fuel placement are the main design problems.

REUSE FROM EXISTING IMPLEMENTS

Frame mounting like the battery pack; the irrigation petrol pump is another engine use.

OPEN QUESTIONS

- Generator power class and mass
- Vibration isolation and noise
- Fuel tank and exhaust placement
- Charging strategy with the battery

Folder: `implement ideas/generator_engine/`

IDEA · POWER & DRIVE

Solar panels

Not worked out yet

WHAT IT DOES

Integrated panels to extend battery life and provide sustainable power during operation.

CONCEPT (FIRST THOUGHTS)

Panels on a flat or slightly tilted roof frame above the top plate, wired through a solar charge controller to the 48 V bus. The roof is also a mounting plane for antennas and cameras, so the GNSS antenna position must stay clear.

REUSE FROM EXISTING IMPLEMENTS

Top-plate pattern of the robot; antenna positions of the other implements.

OPEN QUESTIONS

- Panel area that fits the robot's width
- Energy yield per day versus consumption
- Shadow of implements, mast and antenna
- Wind load and mounting

Folder: `implement ideas/solar_panels/`

IDEA · POWER & DRIVE

Heavy-duty electric motors

Not worked out yet

WHAT IT DOES

Electric motors such as the Ego Power+ or Greenworks units to drive implements that need real power, replacing small petrol engines.

CONCEPT (FIRST THOUGHTS)

A standard interface plate for battery-powered garden-tool motors (their battery or the robot's bus via adapter), with a belt or shaft output to an implement. Gives a ready, cheap drive where a bought motor is easier than a custom 48 V BLDC.

REUSE FROM EXISTING IMPLEMENTS

Motor plates and V-belt drives of the around-pole mower and the weed picker.

OPEN QUESTIONS

- Which motors: brands, power, output shaft type
- Supply: own battery or the robot's 48 V
- Interface plate and quick-change mounting
- Protection against dust and water

Folder: `implement ideas/heavy_duty_motors/`

IDEA · TRANSPORT & GENERAL

Broom / sweeper

Not worked out yet

WHAT IT DOES

Large front-mounted brush or electrically driven radial brush for surface cleaning, as an alternative to the front push blade.

CONCEPT (FIRST THOUGHTS)

A cylindrical (radial) brush across the front with a 48 V motor and height adjustment. The disc brushes already exist for the rear; a roller brush at the front sweeps material sideways or into a collector.

REUSE FROM EXISTING IMPLEMENTS

Disc brushes (brushes/simple): motors, toolbar and mounting; push blade as the alternative.

OPEN QUESTIONS

- Radial brush diameter, length and bristle type
- Sweep direction and collector
- Height and pressure control
- Dust control

Folder: `implement ideas/broom_sweeper/`

IDEA · TRANSPORT & GENERAL

Seat & frame extension

Not worked out yet

WHAT IT DOES

Front or rear-mounted seat, with the option to widen the frame tubes at the sides.

CONCEPT (FIRST THOUGHTS)

A seat on a tube frame so a person can ride on the robot, and side extensions that widen the frame for wider implements. Needs a manual-drive mode and clear operator-safety rules.

REUSE FROM EXISTING IMPLEMENTS

Toolbar and hanger design of the disc brushes; tube sizes from the robot.

OPEN QUESTIONS

- Operator weight and the axle loads
- Controls and an emergency stop within reach
- Side-extension width versus stability
- Safety rules for a ridden robot

Folder: `implement ideas/seat_frame_extension/`

IDEA · TRANSPORT & GENERAL

Dumper / power barrow

Not worked out yet

WHAT IT DOES

Motorized wheelbarrow-style bin for transporting and dumping loose materials, also usable as a carrying aid or mini dumper.

CONCEPT (FIRST THOUGHTS)

A tipping bin on the robot with an actuator, taking loads of loose material to a place and tipping them out. Tip over the front or the rear; gate for controlled discharge.

REUSE FROM EXISTING IMPLEMENTS

Actuator, hanger and frame of the push blade; payload logic like the compost spreader.

OPEN QUESTIONS

- Bin volume and payload
- Tip direction and stability while tipping
- Ground pressure on soft soil

Folder: `implement ideas/dumper_barrow/`

IDEA · TRANSPORT & GENERAL

Tow hitch

Not worked out yet

WHAT IT DOES

Rear-mounted hitch for pulling trailers and other rolling equipment.

CONCEPT (FIRST THOUGHTS)

A drawbar plate bolted to the rear flange holes of the wheel brackets or to the lower beams, with a ball coupling or pin. The powered trailer of the mobile refiller already needs one.

REUSE FROM EXISTING IMPLEMENTS

Trailer coupling of the mobile refiller (liquid) and the flange mounting of the feed pusher auger.

OPEN QUESTIONS

- Maximum drawbar pull and vertical load
- Coupling type
- Reverse driving with a trailer

Folder: `implement ideas/tow_hitch/`

IDEA · TRANSPORT & GENERAL

Counterweights

Not worked out yet

WHAT IT DOES

Ballast weights to improve traction and vehicle balance for heavy tasks.

CONCEPT (FIRST THOUGHTS)

Plates or blocks that clip onto the beams on the 50 mm grid, front and rear, so the centre of gravity can be tuned. Several implements already call for ballast (weed picker, overseeder, liquid applicator v2).

REUSE FROM EXISTING IMPLEMENTS

Ballast notes in the RATIONALE.md of the weed picker, overseeder and applicator.

OPEN QUESTIONS

- Block mass and handling by one person
- Position range front/rear
- Quick-release mounting

Folder: `implement ideas/counterweights/`

IDEA · TRANSPORT & GENERAL

Hydraulic pump

Not worked out yet

WHAT IT DOES

External hydraulic power source for applications where hydraulics are a better option than electricity.

CONCEPT (FIRST THOUGHTS)

A 48 V motor-pump set with a small tank and valve block, with quick couplings at the front and rear for hydraulic implements. Useful for high force at slow speed where linear actuators are too weak.

REUSE FROM EXISTING IMPLEMENTS

Motor and mounting from the heavy-duty motors idea; hydraulics from the irrigation reel (which has its own).

OPEN QUESTIONS

- Flow and pressure class
- Tank size and cooling
- Coupling standard for implements

Folder: `implement ideas/hydraulic_pump/`

IDEA · TRANSPORT & GENERAL

Three-point hitch

Not worked out yet

WHAT IT DOES

Standard lifting system for attaching various agricultural implements.

CONCEPT (FIRST THOUGHTS)

A scaled-down (category 0 or 1) three-point linkage with lift arms, driven by a linear actuator or hydraulic cylinder. It would let standard compact-tractor implements hang on the robot instead of making each lift frame by hand.

REUSE FROM EXISTING IMPLEMENTS

Lift frame with actuator and slotted hole of the liquid fertilizer applicator and overseeder.

OPEN QUESTIONS

- Category 0 or 1, lifting force and stroke
- Rear versus front mounting
- Weight limit imposed by the robot's axle loads
- Electric versus hydraulic lift

Folder: `implement ideas/three_point_hitch/`

IDEA · TRANSPORT & GENERAL

Small cabin

Not worked out yet

WHAT IT DOES

Protective enclosure or housing for the operator.

CONCEPT (FIRST THOUGHTS)

A light cabin or canopy on the frame with a seat. Only meaningful together with the seat and frame extension.

REUSE FROM EXISTING IMPLEMENTS

Seat & frame extension; top-plate pattern.

OPEN QUESTIONS

- Weight and centre of gravity
- Entry and exit, visibility
- Rollover protection

Folder: `implement ideas/small_cabin/`

IDEA · MEASURING & SENSORS

Grass meter / crop scanner

Not worked out yet

WHAT IT DOES

Measuring grass and crop height and health to create task maps for nitrogen or phosphate.

CONCEPT (FIRST THOUGHTS)

A mast or boom with ultrasonic, lidar or a rising-plate meter that records height at each GNSS position, stored as a map. The map becomes a task map for the fertilizer spreader or liquid applicator.

REUSE FROM EXISTING IMPLEMENTS

Camera mast of the spot sprayer; RTK antenna mast of the field marker; logging of the other implements.

OPEN QUESTIONS

- Sensor type: ultrasonic, lidar or plate meter
- Number of sensors across the working width
- Calibration to dry-matter yield
- Map format for the task maps

Folder: `implement ideas/grass_meter/`

IDEA · MEASURING & SENSORS

Soil compaction meter

Not worked out yet

WHAT IT DOES

Measuring and mapping soil density at specific data points.

CONCEPT (FIRST THOUGHTS)

A penetrometer rod pushed into the soil by an actuator at a point, force and depth logged against the GNSS position. The robot stops at each data point (like the field marker).

REUSE FROM EXISTING IMPLEMENTS

Z axis and ground-sensing shroud of the field marker; point list from the marker's coordinate handling.

OPEN QUESTIONS

- Cone size and push force
- Depth range and speed
- Data points per hectare
- Calibration against a manual penetrometer

Folder: `implement ideas/soil_compaction_meter/`

IDEA · MEASURING & SENSORS

NIR sensor (NDVI & crop status)

Not worked out yet

WHAT IT DOES

Near-infrared sensor that measures the NDVI of the crop while driving. From the difference between reflected red and near-infrared light the robot reads biomass, ground cover and crop vitality, and builds live task maps for variable-rate fertilisation. Because the sensor moves close over the crop instead of looking down from a satellite, the readings stay reliable under cloud cover and also register small spots and gaps.

CONCEPT (FIRST THOUGHTS)

Two-band (red and near-infrared) sensor or a multispectral camera on a mast above the crop, with active illumination or a reference panel against sun angle. NDVI is logged per GNSS position and turned into a task map.

REUSE FROM EXISTING IMPLEMENTS

Mast and mounting of the spot sprayer camera; task-map output feeds the fertilizer spreader and liquid applicator.

OPEN QUESTIONS

- Active versus passive sensor, sensor height
- Sun-angle and shadow correction
- Calibration of NDVI to nitrogen need
- Task-map format and update rate

Folder: `implement ideas/nir_sensor/`

IDEA · COOPERATION & SWARM OPERATION

Drone combination

Not worked out yet

WHAT IT DOES

The robot works together with a drone as a mobile base station. The drone flies out from the platform, maps the plot from above and passes its imagery straight to the robot, which converts it into a driving and treatment plan. The drone charges and swaps batteries on the robot, so the pair can keep scouting and working alternately for a full day without anyone coming out to the field.

CONCEPT (FIRST THOUGHTS)

A landing platform on the robot's top plate with a charging contact or battery-swap mechanism, a data link to the robot computer and a way to align the drone on landing (RTK or visual markers). The plan generation reuses the task-map pipeline of the crop scanner.

REUSE FROM EXISTING IMPLEMENTS

Top-plate pattern; solar panel roof as an alternative platform; task-map pipeline of the grass meter and NIR sensor.

OPEN QUESTIONS

- Platform size and landing aid
- Charging versus battery swap
- Weather limits for flying and landing
- Regulations for autonomous flights

Folder: `implement ideas/drone_combination/`

IDEA · COOPERATION & SWARM OPERATION

Swarm operation (multiple robots per field)

Not worked out yet

WHAT IT DOES

Several Bruut units work the same field as one swarm. The job is split into sections, the robots divide the passes among themselves and keep each other informed of position, progress and battery state, so they never overlap or cross. Capacity scales simply with the number of units instead of with machine size, work continues when one unit drops out, and the light platforms keep soil compaction low.

CONCEPT (FIRST THOUGHTS)

Mostly software: a job planner that splits the field into passes, shared position and progress messages over a radio link, collision avoidance by reserved corridors, and hand-over of a pass when one unit drops out. The mobile refiller already works with a working robot, which is a two-unit special case.

REUSE FROM EXISTING IMPLEMENTS

The pass pattern of the mobile refiller (simple) and the headland-turn animations of the agbot slim.

OPEN QUESTIONS

- Communication link and message set
- Pass allocation and re-allocation
- Collision and crossing rules
- Charging and refilling logistics for many units

Folder: `implement ideas/swarm_operation/`